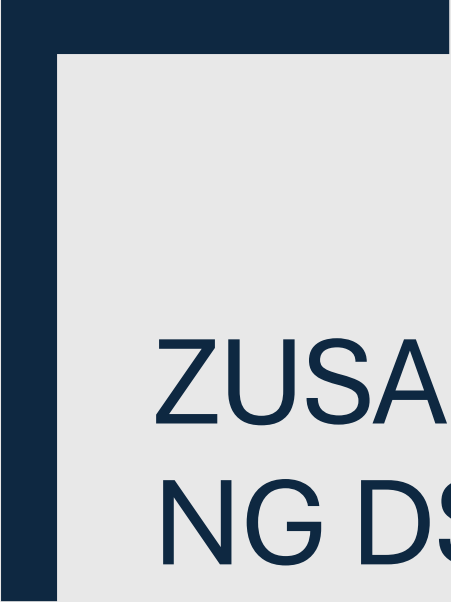




ZUSAMMENFASSU NG DSI 30 THEMEN

Bui Vasin (buiwas01)

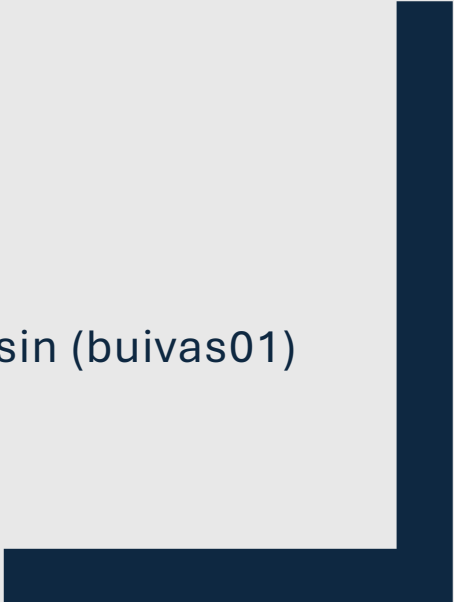


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1. Algorithms, Model Characteristics

Zu den Algorithmen-Typen gehören diese dazu:

Lernarten (Learning-Types): supervised, unsupervised, semi-supervised, reinforcement

Algorithmen: Decision Trees, KNN, SVM, Random Forests, Gradient Boosting, Neural Networks

Anwendungsbeispiele: Klassifikation, Regression, Clustering

Zu den Modelleigenschaften (Model Characteristics) spricht man von:

Overfitting / Underfitting, Bias-Variance Tradeoff, Interpretability, Complexity / Computation Time, Generalization, Model Performance / Evaluation Metrics

A) Algorithm-Types

Eager learners Definition:

Eager learners are machine learning algorithms that first build a model from the training datasets before making any prediction on future datasets.

Eager learners build a general model during training, which is then used to make predictions on new data → They process the entire training data upfront, optimize a hypothesis function, and discard the training data afterward.

Algorithms: Logistic Regression, Support Vector Machine, Decision Trees, Artificial Neural Networks.

Typical characteristics : High training time, fast prediction time, Good generalization if trained properly, Memory-efficient at inference time

Lazy Learners (Instance-Based Learning) Definition :

Lazy learners do not build a model during training. Instead, they store the training data and defer generalization until a query (new data) arrives → The “learning” happens at prediction time.

Algorithms : K-nearest neighbors (KNN) (mainly), Case-Based reasoning, Local regression (aka local polynomial regression, moving regression).

Typical characteristics: Low training time, slow prediction time (because they must search through data), High memory consumption, Sensitive to noisy data & irrelevant features

B) Learning-Types

1. Supervised Learning Definition:

You give the computer a dataset that includes input data and the correct answers (labels). The model learns by looking at examples that say: "This input leads to that output."

Example → A model sees emails labeled as "spam" or "not spam" and learns to detect spam.

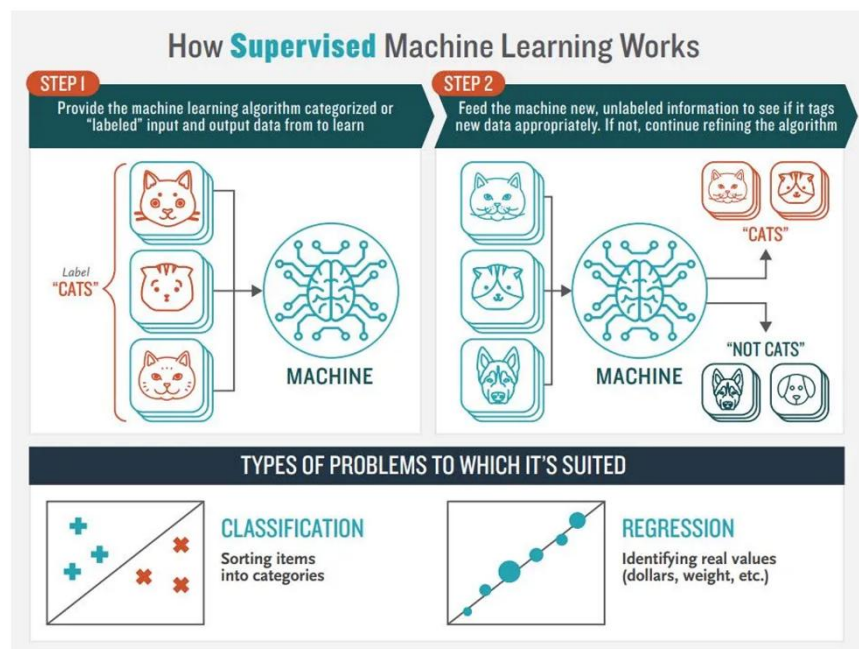
Goal : Learn to predict the correct answer for new data.

Used for : Classification → Is this a cat or a dog in the dataset ?

Regression → What is the price of this House ?

Algorithms : Support Vector Machines (SVM), Neural Networks, k-Nearest Neighbors(KNN), Decision Trees, Linear Regression

Important Key Points : It needs labeled data, Easy to evaluate(you know the right answers), Very common in real-world applications



	ALGORITHM	DESCRIPTION	APPLICATIONS	ADVANTAGES	DISADVANTAGES
Linear Models	Linear Regression	A simple algorithm that models a linear relationship between inputs and a continuous numerical output variable	USE CASES 1. Stock price prediction 2. Predicting housing prices 3. Predicting customer lifetime value	1. Explainable method 2. Interpretable results by its output coefficients 3. Faster to train than other machine learning models	1. Assumes linearity between inputs and output 2. Sensitive to outliers 3. Can underfit with small, high-dimensional data
	Logistic Regression	A simple algorithm that models a linear relationship between inputs and a categorical output (1 or 0)	USE CASES 1. Credit risk score prediction 2. Customer churn prediction	1. Interpretable and explainable 2. Less prone to overfitting when using regularization 3. Applicable for multi-class predictions	1. Assumes linearity between inputs and outputs 2. Can overfit with small, high-dimensional data
	Ridge Regression	Part of the regression family — it penalizes features that have low predictive outcomes by shrinking their coefficients closer to zero. Can be used for classification or regression	USE CASES 1. Predictive maintenance for automobiles 2. Sales revenue prediction	1. Less prone to overfitting 2. Best suited where data suffer from multicollinearity 3. Explainable & interpretable	1. All the predictors are kept in the final model 2. Doesn't perform feature selection
	Lasso Regression	Part of the regression family — it penalizes features that have low predictive outcomes by shrinking their coefficients to zero. Can be used for classification or regression	USE CASES 1. Predicting housing prices 2. Predicting clinical outcomes based on health data	1. Less prone to overfitting 2. Can handle high-dimensional data 3. No need for feature selection	1. Can lead to poor interpretability as it can keep highly correlated variables
Tree-Based Models	Decision Tree	Decision Tree models make decision rules on the features to produce predictions. It can be used for classification or regression	USE CASES 1. Customer churn prediction 2. Credit score modeling 3. Disease prediction	1. Explainable and interpretable 2. Can handle missing values	1. Prone to overfitting 2. Sensitive to outliers
	Random Forests	An ensemble learning method that combines the output of multiple decision trees	USE CASES 1. Credit score modeling 2. Predicting housing prices	1. Reduces overfitting 2. Higher accuracy compared to other models	1. Training complexity can be high 2. Not very interpretable
	Gradient Boosting Regression	Gradient Boosting Regression employs boosting to make predictive models from an ensemble of weak predictive learners	USE CASES 1. Predicting car emissions 2. Predicting ride hailing fare amount	1. Better accuracy compared to other regression models 2. It can handle multicollinearity 3. It can handle non-linear relationships	1. Sensitive to outliers and can therefore cause overfitting 2. Computationally expensive and has high complexity
	XGBoost	Gradient Boosting algorithm that is efficient & flexible. Can be used for both classification and regression tasks	USE CASES 1. Churn prediction 2. Claims processing in insurance	1. Provides accurate results 2. Captures non linear relationships	1. Hyperparameter tuning can be complex 2. Does not perform well on sparse datasets

2. Unsupervised Learning Definition:

Unsupervised Learning works with unlabeled data. The algorithm must discover hidden patterns, groupings, or data structures without prior knowledge of outputs.

You give the computer data with no labels, just raw input. The model must find patterns or groups in the data on its own.

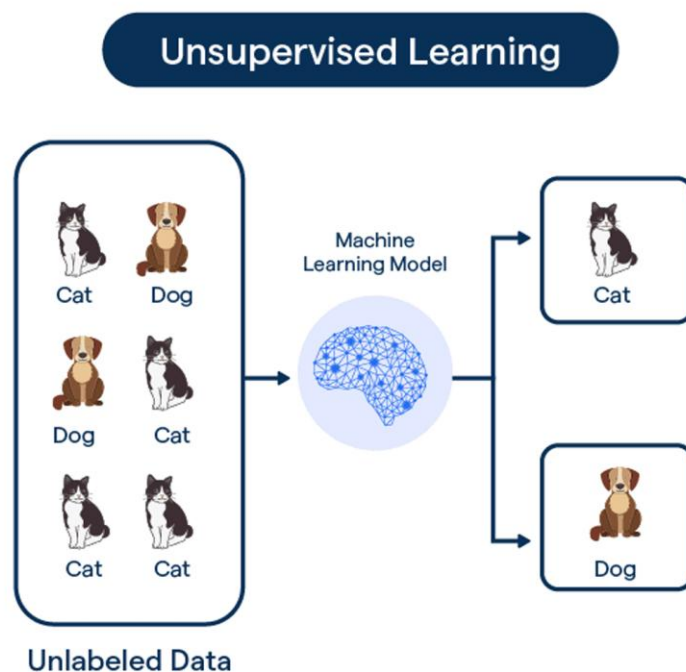
Example: You give the computer a list of customer shopping data, but no categories.

The model finds out: "These people buy similar things – they might be one group."

Used for: Clustering « grouping similar things », Pattern discovery « finding hidden structures »

Algorithms: K-Means, K-Medoids, Fuzzy C-Means etc.

Important Key Points : No labels needed = cheaper data, Harder to evaluate results (no « correct answer »), Great for expolation and pattern finding.



Unsupervised Learning	Clustering	K-Means	K-Means is the most widely used clustering approach—it determines K clusters based on euclidean distances	USE CASES 1. Customer segmentation 2. Recommendation systems	1. Scales to large datasets 2. Simple to implement and interpret 3. Results in tight clusters	1. Requires the expected number of clusters from the beginning 2. Has troubles with varying cluster sizes and densities
		Hierarchical Clustering	A "bottom-up" approach where each data point is treated as its own cluster—and then the closest two clusters are merged together iteratively	USE CASES 1. Fraud detection 2. Document clustering based on similarity	1. There is no need to specify the number of clusters 2. The resulting dendrogram is informative	1. Doesn't always result in the best clustering 2. Not suitable for large datasets due to high complexity
		Gaussian Mixture Models	A probabilistic model for modeling normally distributed clusters within a dataset	USE CASES 1. Customer segmentation 2. Recommendation systems	1. Computes a probability for an observation belonging to a cluster 2. Can identify overlapping clusters 3. More accurate results compared to K-means	1. Requires complex tuning 2. Requires setting the number of expected mixture components or clusters
	Association	Apriori algorithm	Rule based approach that identifies the most frequent itemset in a given dataset where prior knowledge of frequent itemset properties is used	USE CASES 1. Product placements 2. Recommendation engines 3. Promotion optimization	1. Results are intuitive and Interpretable 2. Exhaustive approach as it finds all rules based on the confidence and support	1. Generates many uninteresting itemsets 2. Computationally and memory intensive. 3. Results in many overlapping item sets
		ALGORITHM	DESCRIPTION	APPLICATIONS	ADVANTAGES	DISADVANTAGES

3. Semi-Supervised Learning definition :

A mix between supervised and unsupervised learning. You have a small amount of labeled data, and a large amount of unlabeled data.

The model learns patterns from both – it uses the few labels to better understand the rest.

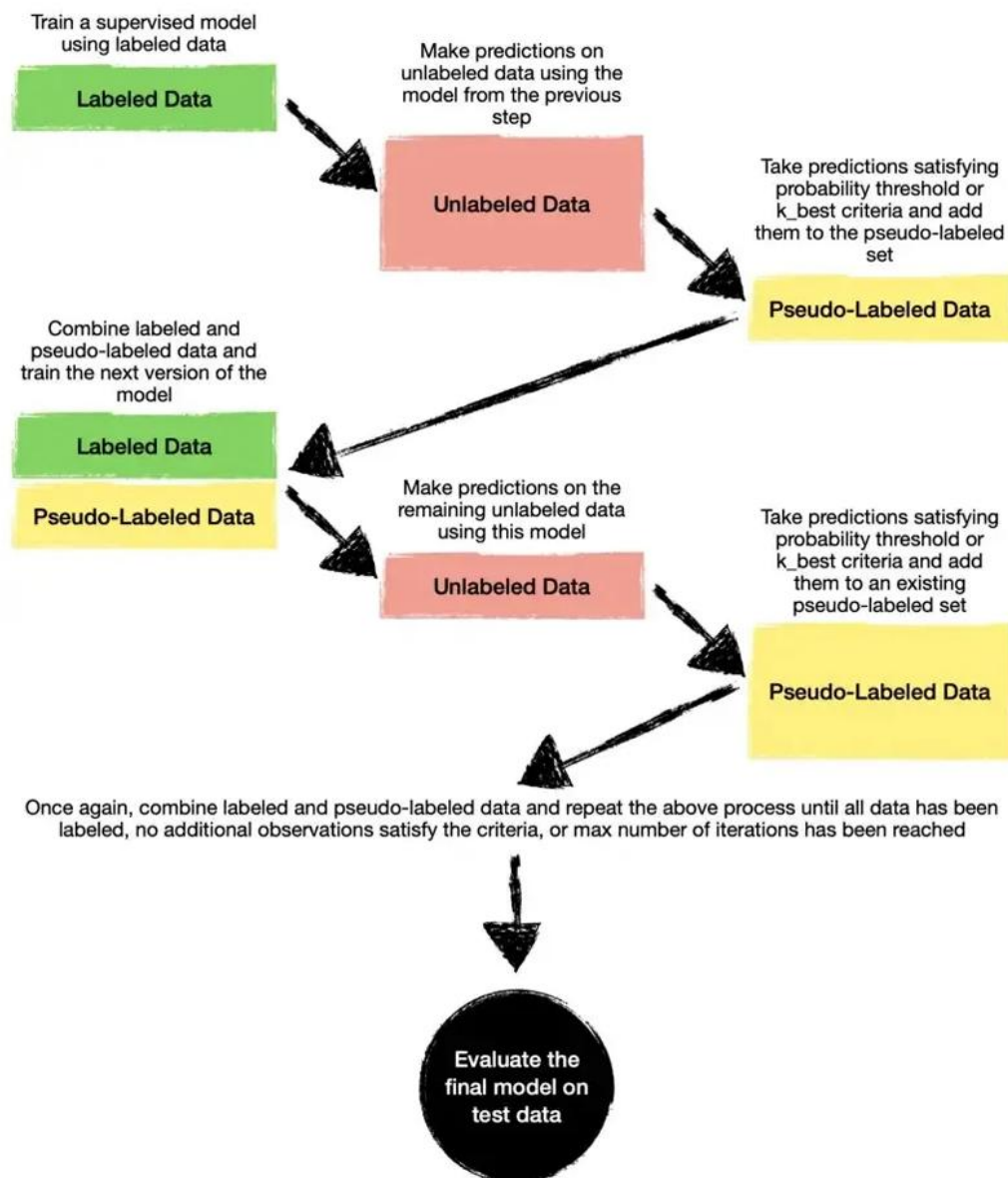
Week 8 Slide 9: “Learning with a small number of labeled examples and a large number of unlabeled examples.”

Example: You label 100 medical images showing tumors. But you have 10,000 more images with no labels.

The model uses the 100 labeled ones to guess what the other images show, and learns faster and cheaper.

Goal: Achieve high performance using very few labeled examples. This helps when labeling is expensive or time-consuming.

Key Points: Saves time and money, Fits real-world situations, Better than unsupervised.



4. Reinforcement Learning (RL) definition :

An agent learns how to behave in an environment by taking actions and receiving rewards or penalties → It improves its behavior through trial and error.

Model-based RL : The agent tries to build a model of the environment – meaning: it learns how the world works (what happens after each action). Once it knows how the environment behaves, it can plan ahead and choose the best actions.

Example: A robot learns that “if I move forward on a slope, I will slide.” It builds a model of how gravity affects it and uses this to make smart choices.

Key Points: Can simulate possible outcomes internally, Often faster learning but more complex to implement, Learns a transition model: “If I do X in state Y, I end up in state Z”

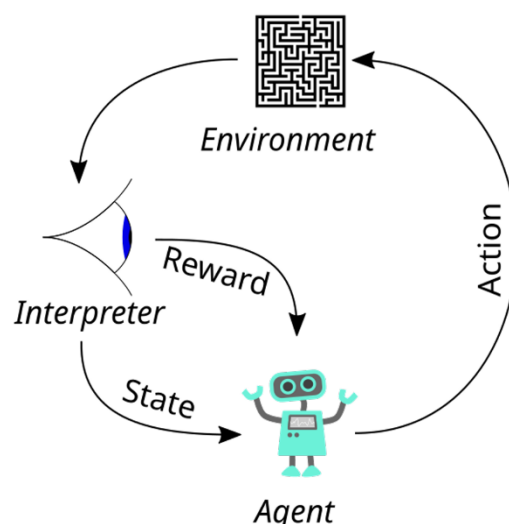
Model-free RL : The agent does not try to model the environment. It just learns by trial and error: take an action → get a reward → try to do better next time. No internal model of how things work – just reaction to outcomes.

Example : An AI plays a game over and over. It doesn't understand the game rules, but slowly figures out what actions help it win.

Key Points : No prediction of what will happen – just learns from actual experience, Simpler but can need more training

Summary :

Type	Builds a model?	Strategy	Example
Model-Based	✓ Yes	Learns how environment works, then plans	Chess-playing robot with strategy
Model-Free	✗ No	Learns only from reward history	Game bot that learns by replaying



5. Zero-Shot Learning definition:

Zero-Shot Learning is a machine learning approach where a model can make predictions about things it has never seen before during training. → The model uses prior knowledge (like word embeddings, concepts, relationships) to generalize to new classes without direct examples. Week 8 Slide 10 : Zero-Shot Learning is listed as one of the learning strategies, related to how a model deals with new, unseen situations without training examples.

Example: You train a model to recognize cats, dogs, and horses.

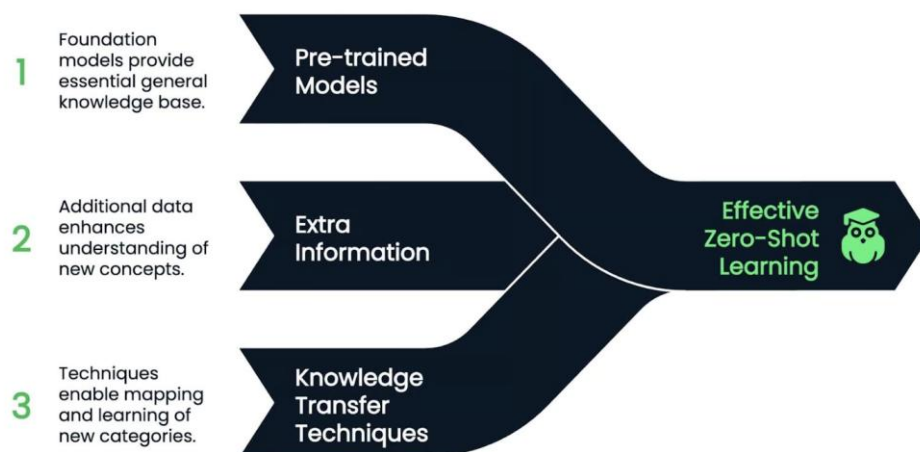
Later, you ask it to recognize a zebra – even though it has never seen a zebra before.

It still succeeds, because it understands that "zebra = horse + stripes" through descriptive features or semantic relations.

Goal : Enable the model to make correct predictions for unseen categories – using descriptions, attributes, or relationships instead of labeled data.

Key Points : Needs a general knowledge model, No labeled training data for the target class, Works well when attributes or descriptions are available.

Building Blocks of Zero-Shot Learning



C) Model Characteristics

In data science, models are abstract representations of reality that learn patterns from data to make predictions or uncover structures. The characteristics of these models determine their performance, generalizability, and reliability.

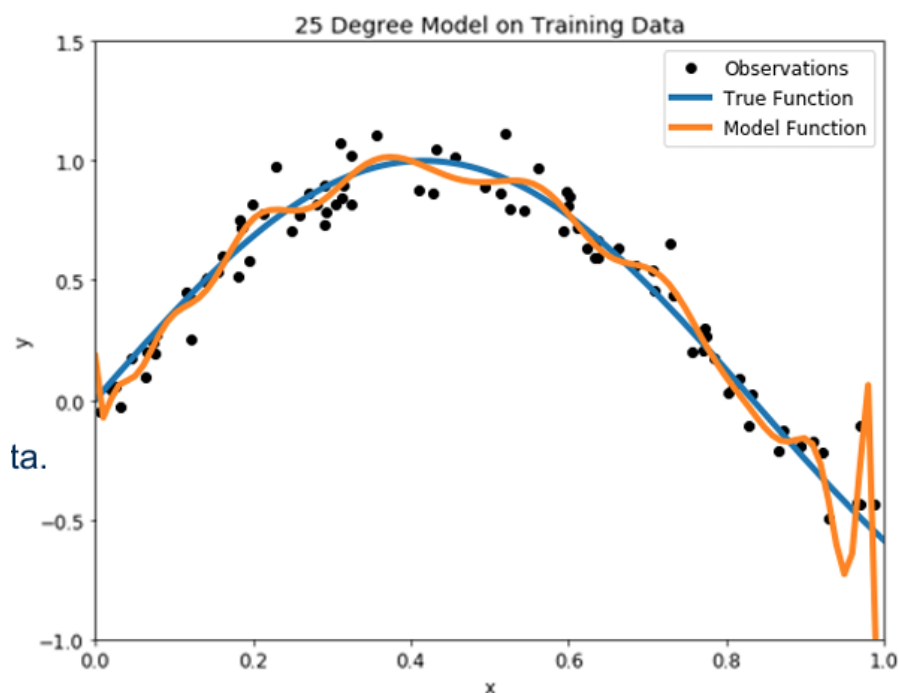
Key Model Characteristics:

1. Model Fit (Underfitting vs. Overfitting)

Overfitting: The model learns the training data too well, including noise and anomalies, leading to poor performance on unseen data. If we fit the (searched) function too closely to the data points, we miss the actual relationship. Consequently, our (searched) function will make poor predictions despite a good fit in the training data. The reason for this is overfitting, which leads to poor robustness.

Key Traits : Performs extremely well on training data, Performs poorly on test data, Learns not only signal, but also noise

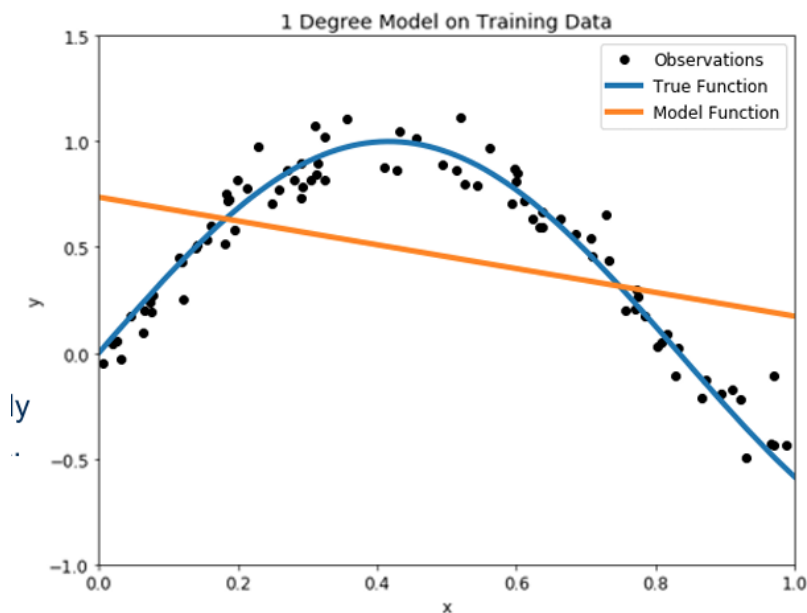
Problem leads to -> Model cannot generalize new data.



2. Underfitting: The model is too simplistic to capture underlying patterns in the data, resulting in high error both on training and test data. If we fit the (searched) function too little to the data points, we also miss the actual correlation. In this case, in addition to the poor predictions in the test data, our (searched) function will already perform poorly in the training data. The reason for this is underfitting.

Key Traits: Performs poorly on training AND test data(important), The model is too simple to capture patterns.

Problem: The model has high bias and misses important relationships.



How to Identify Over-underfitting?

	Underfitting	Overfitting
Training Accuracy	Low	Very high
Test Accuracy	Low	Lower than training, often low
Model Complexity	Too simple	Too complex
Behavior	Fails to capture structure	Memorizes everything

3. Class Imbalance:

A class imbalance exists when the data points per class are not present in equal amounts. Occurs when certain classes are significantly underrepresented compared to others in the dataset.

Key Points: Leads to misleading accuracy (model may always predict majority class), Minority class may be ignored by the model, Examples: fraud detection, rare disease detection.

Solutions : Undersampling: Reduce majority class size, Oversampling: Duplicate or synthetically generate minority data, SMOTE: Synthetic Minority Over-sampling Technique, Class weighting in model training.

What does Undersampling/Oversampling/SMOTE mean ?:

Undersampling : Reducing the number of examples from the majority class so that both classes are more balanced. (800 vs 100 samples to 100 vs 100).

Oversampling : Increasing the number of examples from the minority class by duplicating or generating new samples (800 vs 100 to 800 vs 800 samples).

SMOTE(Synthetic Minority Over-sampling Technique): Instead of duplicating minority samples, SMOTE creates new, synthetic samples by interpolating between existing ones. Class Weighting : You tell the algorithm to give more importance to the minority class when calculating loss during training.

4. Bias-Variance-Tradeoff definition :

Describes the tension between bias (underfitting) and variance (overfitting). A good model balances both. (It's a fight you want them to be balanced not too much bias and not too much Variance).

Model Behavior	Explanation
High Bias	Too simple → underfitting
High Variance	Too complex → overfitting
Balanced (ideal)	Good generalization

Key Points : Fundamental concept in model evaluation, Goal: find the “sweet spot” in complexity

2. CPS(Cyber-Physical-Systems) Integration

Definition :

A Cyber-Physical System is a system that tightly integrates computational (cyber) elements with physical processes, usually over a network. It collects data from the real world via sensors, processes this data using software, and performs actions via actuators.

IMPORTANT!: A CPS = sensors + software + connectivity, “When is something cyber?” → When it processes, decides, and interacts dynamically with the physical world

They collect, process, and act upon data from the real world in real-time.

CPS examples: smart grids, autonomous vehicles, connected factories, smart homes.

Key Features of CPS :

Feature	Description
Sensor-based	Measures real-world variables (e.g., speed, temperature, movement)
Networked	Communicates over local or global networks (IoT, 5G)
Real-time capable	Makes decisions and reacts in milliseconds
Feedback loop	Acts, observes the result, and adapts (control systems)
AI/ML enhanced	Learns from patterns to optimize future behavior

CPS vs IoT vs Embedded Systems

System Type	Primary Focus	Intelligence	Example
Embedded System	Local device control	✗ Low	Microwave controller
IoT Device	Data connection + remote access	✗ Limited	Smart plug
CPS	Intelligent feedback & control	✓ High	Self-driving car, Smart Grid

The Role of Data Science in CPS :

CPS are massive data producers, and Data Science enables :

Predictive Analytics – Predict failure before it happens (e.g. in turbines, engines)

Anomaly Detection – Detect unusual sensor readings

Machine Learning Integration – Systems that adapt (e.g., Reinforcement Learning)

Control Optimization – Adjusting system behavior automatically

Stream Data Processing – Real-time analysis of sensor input

Why CPS integration Matters :

CPS form the technological backbone of:

Industry 4.0, Smart Cities, Autonomous Vehicles, Predictive Healthcare, Digital
Factories,

CPS enable :

Real-time decision-making, Continuous feedback and improvement, Intelligence at the
edge (near the sensors)

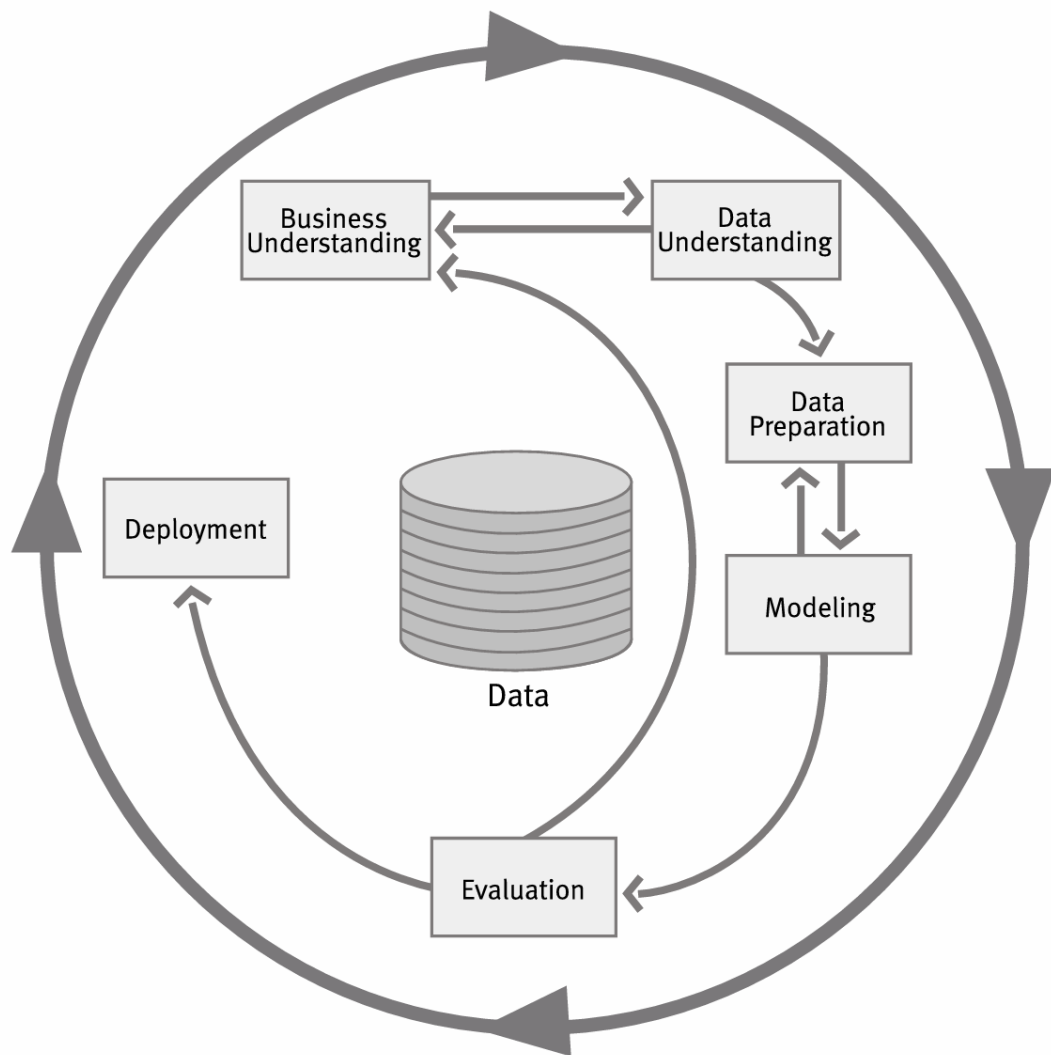
3. Problem Solving Evolution

Problem framing is a critical step in data science that ensures the correct problem is being addressed. Defining the problem properly influences model selection, data collection, and the overall success of a project. Misframing a problem can lead to incorrect insights and decision-making failures.

Persona : The Data Scientists (Hitz Frage)

Data Scientists work at the intersection of statistics, programming, and business understanding. A challenge in large organizations is the "throwing over the wall" phenomenon, where 87% of data science projects fail to reach production. There are a lot of different reasons for that, for instance the model produced useless results, or the domain knowledge was simply not captured correctly
«All models are wrong, but some are useful» - George Box.

CRISP-DM(Cross-Industry Standard Process for Data Mining)



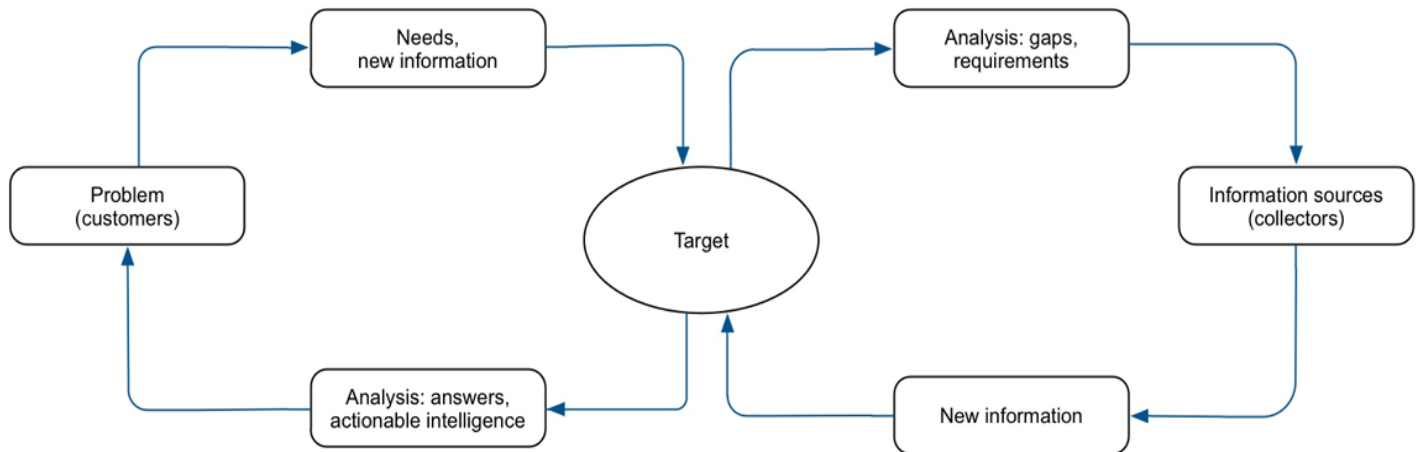
A widely used framework for data mining projects, consisting of:

- Business Understanding
- Data Understanding
- Data Preparation
- Modeling
- Evaluation
- Deployment

*Important note: If the problem is not framed correctly the result will be useless.

Target-Centric approach:

The Target-Centric Approach is a structured, iterative method for gathering and analysing information to enhance decision-making. Unlike linear models, it continuously refines insights as new data emerges, ensuring a dynamic and adaptable analysis process.



System Dynamics Overview :

Key concepts:

- **Feedback loops** – Reinforcing (positive) or balancing (negative) mechanisms.
- **Delays** – Time gaps between cause and effect.
- **Accumulations** – Stocks that change over time due to inflows and outflows.

Key System Dynamics Concepts

Control Loop Model: Used for management systems to maintain stability in operations.

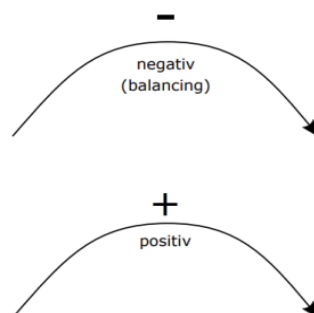
- **Relationship Types:**

Antagonal Relationship:

- the bigger ... the smaller
- the smaller ... the larger
- the less ... the more

Equidirectional relationship:

- the greater ... the greater
- the stronger ... the stronger
- the smaller ... the smaller
- the less ... the less



Stock-Flow Diagrams: Model systems using:

- **Stocks** (accumulations)
- **Flows** (inputs and outputs)
- **Parameters** (influencing variables)

Importance of Problem Framing in Data Science

Problem framing is a critical step in Data Science, as it defines the scope, objectives, and constraints of a project. A well-structured problem statement helps in selecting the right models, methodologies, and data sources, ensuring that the analysis aligns with the actual business or operational need. When a problem is not properly framed, it can lead to misleading insights, inefficient decision-making, and project failures.

Creativity Techniques

Various methods help in defining and structuring problems:

- Mind Mapping: Visualizing problem components.

- Brainstorming & Brainwriting: Generating ideas individually or in groups.

- Nominal Group Technique (NGT): A structured method for ranking ideas.

- Delphi Method: Expert-based iterative forecasting.

- Design Thinking: User-centric problem-solving.

Nominal Group Technique (NGT)

Steps:

1. Collect ideas individually.
2. Write contributions on a flipchart.
3. Discuss ideas.
4. Select key ideas.
5. Vote on the best solutions.

4. VUCA Decision-Making

What is VUCA ?

Volatility – rapid, unpredictable change

Uncertainty – lack of predictability, limited foresight

Complexity – many interdependent variables

Ambiguity – unclear meanings, multiple interpretations

« First introduced by the US military, now essential in business & AI contexts.»

A Guide to Approaching Events in the Four VUCA Categories



Why does VUCA matter in Data Science ?

Modern data science works in:

Rapidly changing environments (volatility)

Incomplete, noisy, and biased data (uncertainty)

High-dimensional problems (complexity)

Ill-defined goals and relations (ambiguity)

In a VUCA world, simple models and fixed strategies fail.

We need adaptive, systemic, and feedback-based decision-making.

“Only organizations that sense (data), think (models), and act (policies) in one closed loop can thrive.”

The Evolution of VUCA (ZHAW Slide)

Timeframe	VUCA Drivers
1960s	Oil crises, Cold War, linear industrial systems
2020s	Digital platforms, AI, pandemics, misinformation

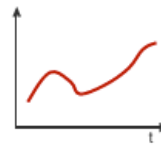
How to deal with VUCA? → System Thinking

System Dynamics allows us to model and reason within VUCA conditions using:

Tool	Purpose
Causal Loop Diagrams	Show feedback effects (+/- loops)
Stock and Flow Diagrams	Track accumulations and transitions
Mental Models	Capture how individuals view the system
Shared Models	Build consensus understanding (decision alignment)
«A model is a hypothesis with a steering wheel» -> From Episode 6 Flight Simulators	Test decisions in virtual environments (safe-to-fail)

PHAPI Loop – Data Driven Decision Cycle

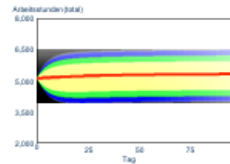
P'roblem
(“reference mode”)



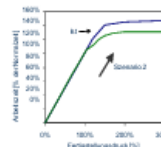
Hypotheses
(= the formal model)



Analyses
(simulation experiments)



Policy design
(testing improved structures)



Implementation



Fig: PHAPI - System dynamics process (J. W. Forrester, 1994, p. 245)

Step	Description
P – Problem	Define system state to observe (e.g., customer churn)
H – Hypothesis	Form ideas on what drives the system behavior
A – Analysis	Use data science tools to simulate/test models
P – Policy	Translate insights into concrete actions or decisions
I – Implementation	Deploy solution, monitor effectiveness (e.g., A/B tests, drift)

PHAPI closes the loop from data to insight to action

5. Human-Centric AI

What is Human-Centric AI?

Human-Centric AI is the approach to designing and applying artificial intelligence systems that:

Serve human needs and values

Respect human rights and autonomy

Include human oversight, transparency, and fairness

It basically means that : »AI should help people, not replace or harm them«

OECD AI Principles :

Applies to AI and Data Science:

- Inclusive growth and well-being
- Human-centered values
- Transparency and explainability
- Robustness and safety
- Accountability

Normal en scheiss Code (ACM Code of Ethics) :

From the Association for Computing Machinery:

- Contribute to society and human well-being
- Avoid harm
- Be honest and trustworthy
- Respect privacy and confidentiality

10 Principles of the Ethical Use of AI

#	Principle	Description
1	Do no harm	Avoid harm to individuals, society, environment
2	Purpose, necessity, proportionality	AI use must be justified and not excessive
3	Safety & security	Minimize risks, ensure system robustness
4	Fairness & non-discrimination	Prevent bias; ensure equality
5	Sustainability	Support long-term social and environmental well-being
6	Privacy & data governance	Respect GDPR, ensure data protection
7	Human autonomy & oversight	Humans remain in control of key decisions
8	Transparency & explainability	Users can understand how AI works
9	Responsibility & accountability	Assign liability and auditability
10	Inclusiveness & participation	Engage diverse stakeholders and communities

EU Ethics Guidelines for Trustworthy AI : Seven Key Requirements

- Humanagency and oversight
- Technical robustness and safety
- Privacy and data governance
- Transparency
- Diversity, non-discrimination, and fairness
- Societal and environmental well-being
- Accountability

HITL(Human in the Loop)

Considerations:

- Does a human review critical AI outputs?
- Can the system be overridden by humans in case of error?
- Mechanisms for user feedback / error correction?
- Is the AI decision-making process transparent to end-users?

«Human-in-the-loop » ensures humans can:

- Oversee decisions
- Override incorrect or unethical outputs
- Audit and retrain systems

Explainability (XAI)

XAI presents a set of techniques and methods to understand why AI makes a particular decision. Knowledge of the reasons for a decision is considered key to understanding model decisions and improving users' trust in AI-based solutions.

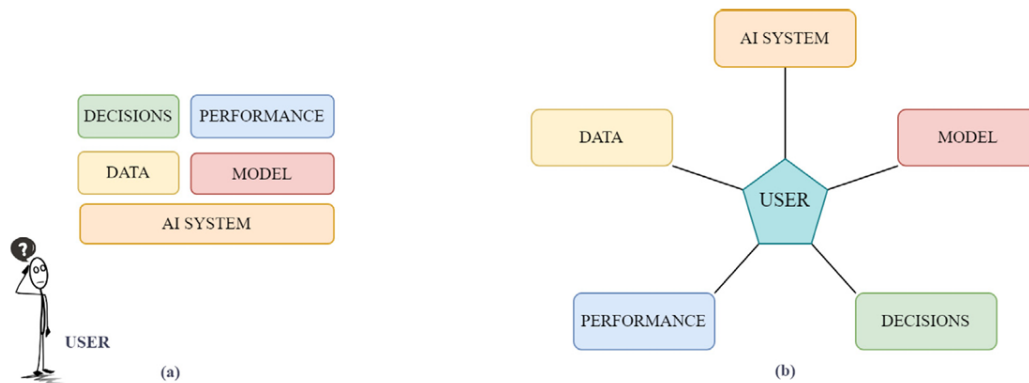


Fig: Design of two explainability approaches in AI: (a) a conventional approach that does not consider the different user groups when presenting explanations, and (b) a user-centric approach that provides explanations tailored to the requirements of different user groups. (Kaplan et al., 2024, p. 5)

SYSTEM PANEL	AI SYSTEM INFORMATION		
	- Overview/Summary of AI System/application - What it is? - Why is it built for? (purpose of the system) - How is it used?		
	DATA DETAILS		
DATA PANEL	<u>Source Info</u>	<u>Exploratory Stats</u>	<u>Data onboarding</u>
	- license - nature of data (modality, domain) - data collection - data annotation - case/class info - sample visualizations - data version	- data distribution - train/validation/test data stats - outlier information - summary stats	- data distribution - train/test data stats - pre-processing steps
MODEL PANEL	MODEL DETAILS		
	<u>Model Info</u>	<u>Development</u>	<u>Deployment</u>
	- model type - model architecture - model visualization - model input and outputs - model behavior - model license	- training framework - hyperparameters - hardware details - reproducibility and code reuse	- framework - model input output details - mode size - inference time

PERFORMANCE PANEL	EVALUATION DETAILS		
	<u>Evaluation Metrics</u>	<u>Performance summary</u>	<u>Limitations</u>
	- details of evaluation metrics - information about test/validation set	- performance details on test/validation set - model performance visualization (expected failure and success)	- limitations regarding the performance and possible error cases
DECISION EXPLANATION PANEL	DECISION EXPLANATION and EXPLORATION		
	<u>Global Explanations</u>	<u>Instance Explanations</u>	<u>Robustness Tests</u>
	- representative cases - borderline cases - data manifold visualization	- instance level post-hoc explanations - comparison analysis with influencer samples from gallery set	- instance manipulation for sensitivity analysis - adversarial attacks - synthetic samples by deep generative models - decision review and correction for further model development

6. Feature Engineering

Definition :

Feature Engineering is the process of transforming, creating, and selecting variables (features) from raw data to improve model performance, interpretability, and robustness. It bridges the gap between raw data and the model's ability to learn.

Why it Matters:

Good features help the model learn patterns better and faster.

Poor features introduce noise, overfitting, or bias.

It often has more impact than the choice of algorithm itself.

“Better features beat more complex models.»

Ensemble Techniques in Machine Learning

Goal: Higher Accuracy, better Performance

Core Idea:

Ensemble learning involves assembling multiple models, such as:

- regression,
- KNN,
- decision trees, and
- clustering

into a cohesive unit. The core idea is to leverage diverse algorithms or distinct datasets with the same algorithm to achieve a more reliable and accurate prediction. The collective intelligence of multiple models often outshines individual predictions (Gandhi, 2024).

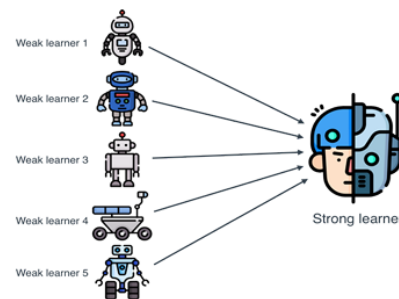


Fig: Ensemble Techniques in Machine Learning (Gandhi, 2024)

Feature Engineering in ML (Slide 27 Week9)

Feature Engineering means preparing and improving data so that machine learning models can learn better and make more accurate predictions.

Understand the Data

Look at your data carefully: What kind of values do you have? Are there missing values or strange numbers?

1. Handle Missing Values

If data is missing, you can either fill it in (like using the average) or remove the row or column.

2. Convert Categories to Numbers

Machine learning needs numbers. Change words like “red”, “blue” or “yes”, “no” into numbers.

3. Use Feature Vectors

Each row of data becomes a list of numbers (a vector) that the model uses to learn.

4. Select the Important Features

Not all features help the model. Keep only the useful ones to make the model simpler and better.

5. Scale the Numbers

Make sure all numbers are in the same range, so no feature dominates others.

6. Create New Features

Make new features from existing ones, like combining “price × quantity” to get “revenue”.

7. Handle Outliers

Very big or small values can confuse the model. Fix or remove them.

8. Use Time Information

If you have time data, create features like “day of week” or “value 1 day ago”.

9. Normalize the Data

Put all numbers into a fixed range (like 0 to 1) so models can train better.

10. Group Numbers into Categories (Binning)

Turn continuous numbers into groups, like “young”, “middle-aged”, “old”.

11. Prepare Text Data

Clean up text: split it into words, remove common words, and reduce to basic forms.

12. Avoid Data Leakage

Don't use information from the test data during training — it will give fake good results.

13. Use Cross-Validation Correctly

Do all feature steps inside the training set during testing. If not, the evaluation will be wrong.

Data Leakage in ML:

Data leakage happens when a machine learning model uses information it should not have during training — for example, test data or future data.

This means the model learns something it wouldn't know in real life.

That makes the model look better during training, but it fails when used on new data.

This problem is called overfitting, and it leads to bad real-world performance.

Why is it a Problem ?

The model "cheats" by using hidden answers.

It learns from data that was supposed to stay secret (like the test set).

It gives fake-good results in training but performs badly in real use.

Examples of where leakage can happen (from the images on the slide):

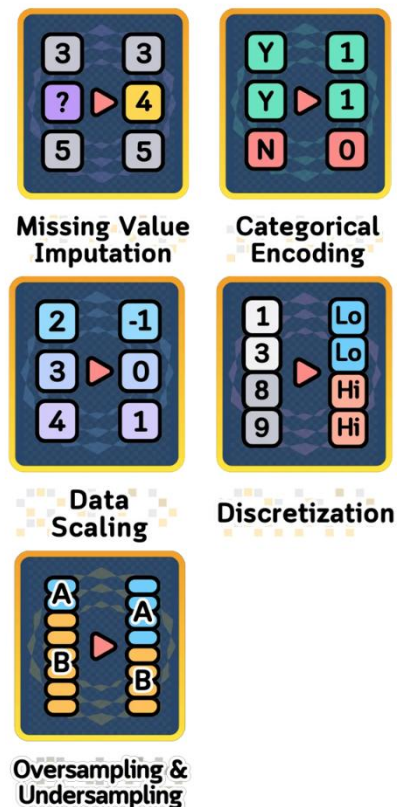


Fig: Data Leakage in Machine Learning (Baladram, 2024)

Missing Value Imputation

If you fill in missing values using the entire dataset before splitting into train/test, you leak info from the test set.

Categorical Encoding

If you encode categories (like "Yes"/"No") using the full data — the test data influences the model.

Data Scaling

If you scale numbers (e.g., normalize) before splitting the data, the scaling includes test info.

Discretization (Binning)

Grouping continuous values into ranges before splitting also leaks test data into training.

Oversampling & Undersampling

If you balance classes (like duplicating minority examples) on the full dataset, test data gets reused in training.

Common Feature Types

In machine learning, a feature is a piece of data (a column in your dataset) that the model uses to learn and make predictions. There are different types of features, and each one needs special handling.

Numerical Features

These are normal numbers like age, height, or salary. You can do math with them (add, subtract, etc.).

Example: 25, 175.5, 3200

Categorical Features

These are labels with a fixed set of options. The model doesn't see them as numbers, just as different groups.

Example: gender (male, female), color (red, green, blue)

Ordinal Features

These are categories that have a clear order or ranking.

The order matters, but the distance between them isn't measured.

Example: T-shirt size (S < M < L < XL), skill level (beginner < intermediate < expert)

Binary Features

A special type of category with only two choices — like yes or no.

It's simple: 0 or 1.

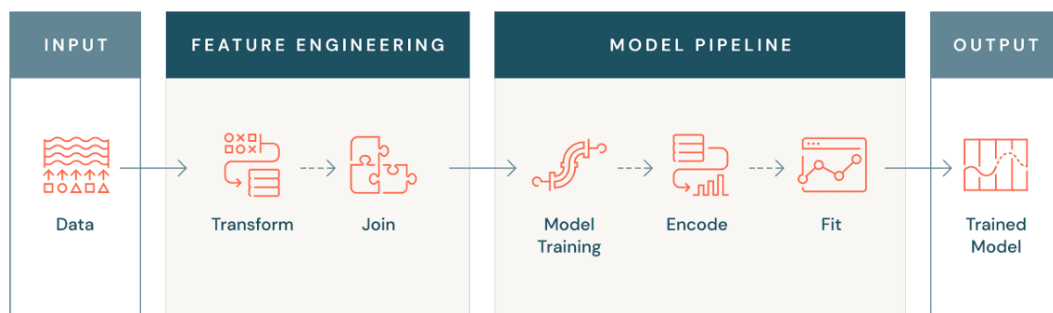
Example: is_smoker (yes/no), has_subscription (true/false)

Text Features

These are features made of words or full text.

They need to be cleaned and broken down into parts using techniques like tokenization before they can be used.

Example: "This product is great!"



The diagram shows the workflow:

Input: Raw data : Feature Engineering: Transform and combine features ,
Model Pipeline: Train and fit the model ,Output: A trained model ready to
make predictions

7. Interdisciplinary Roots of Data Science

Topic 7 is about where Data Science comes from and how it uses ideas and methods from different fields. Think of it like building something with pieces from many different toolboxes. This topic is looked at as part of "Data Science Thinking". The sources show Data Science as something that mixes ideas from many different areas. This is called being "transdisciplinary". There is a picture that helps show the main areas that Data Science uses:

Math, Statistics, Computer Science, Information Technology (IT), Knowledge about the specific area you are working in (Domain Knowledge), Social Science

Mixing these areas is needed to solve problems that cross different fields and to create solutions that also mix ideas from these fields.

What's important to learn about this topic, based on the sources, is understanding these different areas and why Data Science needs all of them:

Understanding what being "transdisciplinary" means for Data Science. It's not just combining things, but really bringing them together to solve complicated problems.

The sources suggest this is needed because the world is complex (like the VUCA world - Volatile, Uncertain, Complex, Ambiguous) and data is getting bigger and more varied (Big Data).

Knowing the main areas Data Science gets its ideas from. The sources point out these areas are:

Math and Statistics: These give the basic tools for understanding and working with numbers and data. They help you summarize data, find patterns, and test ideas.

Computer Science and IT: These provide the skills and tools needed to handle large amounts of data, create and use computer programs (algorithms), and manage data systems. Things like dealing with Big Data (Volume, Variety, Velocity, Veracity) and building systems like Data Warehouses or Data Lakes come from here.

Domain Knowledge: This means knowing a lot about the specific topic or business area you are trying to solve a problem in. You need this to understand the problem, find the right data, make sense of the results, and figure out what actions to take. Not including this can be a problem, sometimes called "solitary analysis".

Social Science: This area helps us think about how Data Science affects people and society. It includes looking at ethical issues like fairness, making sure people are responsible (accountability), and being open about how things work (transparency). It also involves thinking about how humans and AI work together (Human-Centric AI, Human-in-the-Loop).

Understanding why this mix of areas (interdisciplinarity) is key for "Data Science Thinking". This way of thinking means you need more than just technical skills. You need to look at problems from different angles, use scientific methods, and understand the bigger picture of your work.

In short, to learn this topic well, you need to understand that Data Science is built on ideas from many different fields, from math and technology to knowing the problem area and thinking about people and society. This broad base is what allows Data Science to tackle complex problems in the real world (Short Topic).

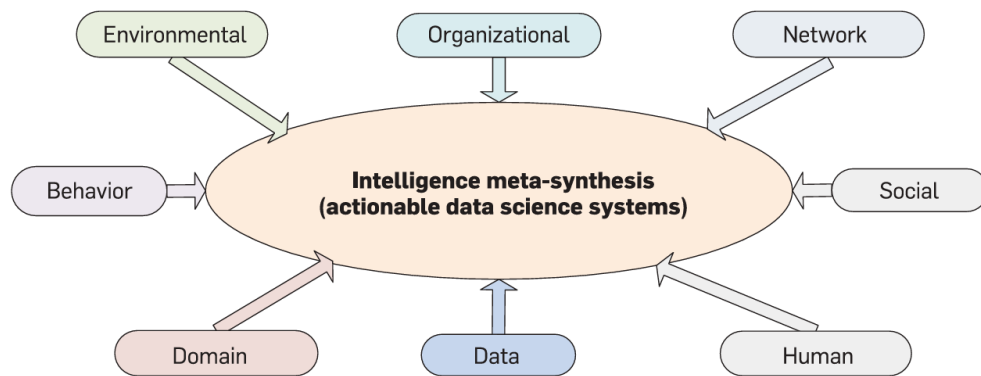


Fig: Synthesizing X-intelligence in data science (Cao, 2017, 2018, pp. 66, 135)

What are the sources of intelligence?

All these areas bring important insights to the system:

1. Environmental – Conditions and trends in the outside world (e.g. climate, regulations, resources)
2. Organizational – Structures, rules, and goals inside companies or institutions
3. Network – Information from communication, connections, or systems
4. Social – People's behavior, opinions, group dynamics
5. Human – Human expertise, intuition, decisions, values
6. Data – The raw data itself (from sensors, users, systems...)
7. Domain – Specific expert knowledge in a field (e.g., medicine, finance)
8. Behavior – Observed patterns in how people or systems act over time

Key Ideas :

Good data science is not just about algorithms — it's about combining human, social, technical, and contextual knowledge to make smarter systems.

This is what Cao calls “X-intelligence” – intelligence that comes from multiple sources and is synthesized into a unified system.

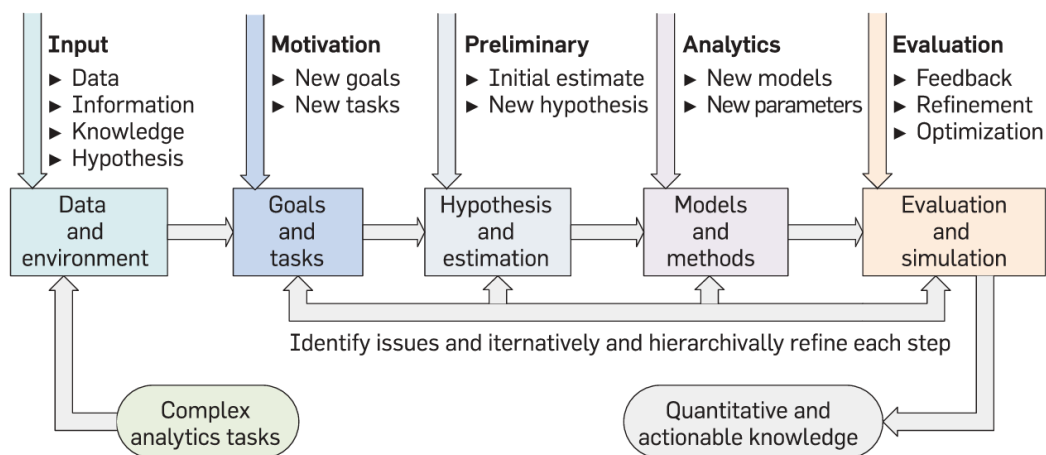


Fig: Complex data science problems: qualitative-to-quantitative X-intelligence (Cao, 2017, 2018, pp. 66, 138)

This flowchart explains how to solve complex data science problems step by step. It combines qualitative thinking (like ideas and goals) with quantitative analysis (like data and models). Each step feeds into the next, and the process loops back to improve results over time.

Main Steps in Simple Terms

1. **Input**
You start with raw material:
→ Data, information, expert knowledge, and first ideas (hypotheses)
2. **Data and Environment**
Understand where the data comes from and what it represents in the real world.
3. **Goals and Tasks**
Define the main goal: What do you want to achieve? What is the problem?
4. **Hypothesis and Estimation**
Make a first guess (hypothesis) based on what you know, and estimate what the data might show.
5. **Models and Methods**
Build and test machine learning models or analysis methods to solve the task.
6. **Evaluation and Simulation**
Check how good your model is. Use feedback to improve and optimize it.

Iterative Process

- After each step, you go back and improve the earlier steps.
- The process is not linear – you refine things as you learn more.
- The goal is to go from ideas to working, useful models.

Turn complex real-world problems into quantitative and actionable knowledge — things you can measure, understand, and act on.

8. Human in the Loop (HITL)

What is Human-in-the-Loop (HITL)?

Human-in-the-Loop means that a person is involved in the process of an AI system. It is not a system that works completely alone (fully autonomous).

The human can help with making decisions, watching the system, or correcting mistakes the AI makes.

This idea is a key part of designing AI systems with a "Human-centric design", meaning focusing on people.

Why is Human-in-the-Loop Important?

Trust: When humans are involved, people often trust the AI system more.

Ethics (Doing the Right Thing): Important rules for using AI safely and fairly require that humans stay in control and are not overruled by AI.

The UN Principles for the Ethical Use of Artificial Intelligence state that AI should not take away human freedom or control, and human oversight must be guaranteed.

Decisions about life or death or basic human rights should never be left only to AI.

The EU's Ethics Guidelines for Trustworthy AI list "Human agency and oversight" as the first important rule.

Reducing Problems: Humans can help find and fix problems like biased results or wrong decisions made by the AI. This can also help stop "Automation Bias", which is when people trust automated results too much, even if they are wrong.

Responsibility: Ethical rules connect HITL to the question of who is responsible when an AI system makes a mistake.

Someone needs to be accountable.

Learning and Improving: When humans give feedback or correct errors, the AI system and the people who made it can learn and get better.

How AI Affects Humans: We also look at how AI changes how humans think and make decisions. Does the AI help human intelligence (like giving them new tools or information)? Does it change how much people trust or rely on the AI?

How to Check for Human-in-the-Loop (Practical Framework):

When looking at an AI system, there are specific things to check regarding HITL, especially in Step 5 of the "Practical Framework to Assess AI Systems".

Are important decisions from the AI checked by a person?

Can a human take over or correct the system if it makes a mistake?

Can users give feedback or report errors?

Is it clear to the people using the system how the AI made its decision? (Is it transparent?)

Also, in Step 6 of the framework, related to HITL, we ask:

Does the AI make human intelligence better?

How does the AI change how much users trust and rely on it?

Are users taught how to understand the AI's results correctly?

Key Things to Learn about HITL:

To understand this topic well, remember:

The definition of HITL and why humans are part of the AI process.

The main reasons why HITL is necessary (ethics, trust, safety, fixing mistakes, responsibility)

The specific questions from the Practical Framework (Steps 5 and 6) that help evaluate if an AI system has good HITL,.

How HITL is connected to important ethical rules like those from the UN and the EU,.

How HITL helps deal with problems like bias and trusting automated systems too much.

In short, Human-in-the-Loop is a crucial idea for making sure AI systems are safe, fair, and work well together with people.

9. Target Centric Farming

Definition: Target-Centric Framing means that we build and guide a data science project by focusing clearly on the main goal (the “target”) from the very beginning.

Instead of starting with data or tools, we ask:

“What is the problem we want to solve?”

“What outcome do we need?”

It’s about keeping the end goal in focus at every step.

Key Concepts:

Start with the target: Always define what you are trying to achieve, Example: “Predict customer churn” or “Detect fraud”.

Design everything around the target: Choose your data, features, methods, and metrics based on what helps achieve that goal, Avoid just “playing with data” without purpose.

Make goals measurable: Your target should be clear and quantifiable, Example: “Reduce churn by 10% in 3 months”

Align stakeholders: Everyone (data team, business, domain experts) should agree on the same target, Avoid confusion or drifting focus.

Avoid data-first mistakes: Many projects fail because they start with available data, not the real need, Target-centric framing helps avoid wasted effort.

Real-world Example

✗ Data-first approach:

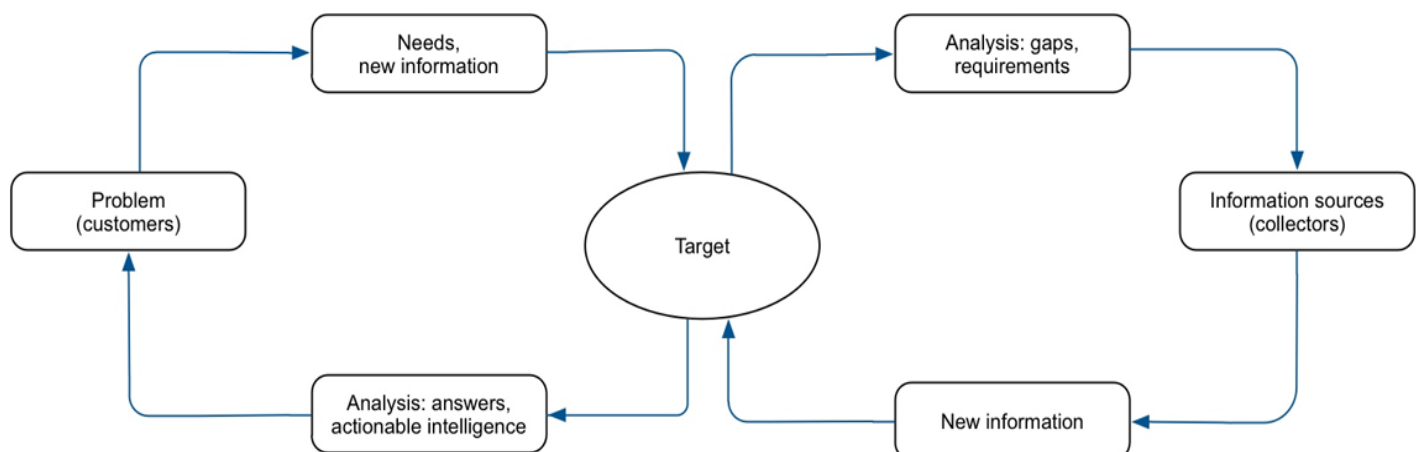
“We have this data, what can we do with it?”

✓ Target-centric approach:

“We want to reduce late deliveries. What data do we need to solve that?”

Target-centric framing means: “Start with the goal – design everything else around it.”

It helps keep the project focused, relevant, and successful. (Btw das Gleiche wie Target Centric Approach)



10. VUCA Role Limits

refers to the boundaries or constraints [response to previous query] that traditional methods, systems, or organizations encounter when the complexity of the environment exceeds their own capabilities.

Here are the key points about this concept:

Context in the VUCA World: This concept is intrinsically linked to the VUCA world, characterized by Volatility, Uncertainty, Complexity, and Ambiguity. While Topic 4 discusses decision-making challenges and Topic 14 covers broader challenges in this environment, Topic 10 specifically focuses on the points at which these challenges cause existing capabilities or systems to reach their limits or become overwhelmed.

Connection to Ashby's Law: The concept of "VUCA Role Limits" is directly related to Ashby's Law of Requisite Variety. This law states that "When the variety or complexity of the environment exceeds the capacity of a system (natural or artificial) the environment will dominate and ultimately destroy that system". The "VUCA Role Limits" are therefore encountered when the variety or complexity inherent in the VUCA environment surpasses the system's ability to manage it.

Consequences: When systems or organizations hit their "Role Limits" in a VUCA environment, rigid, traditional approaches can fail. This can lead to what is sometimes referred to as "fatal undercomplexity", meaning that the system's own control mechanisms are insufficient to cope with the complexity of the situation. The variability and diversity of the environment become too great for the system's steering capacity.

Relevance for Data Science: The necessity for Data Science and data-driven approaches (as discussed in Topic 6 and related to Topic 4) arises partly because conventional methods confront these "VUCA Role Limits. Data Science can help to better understand the variety of the environment and increase one's own capacity to handle this complexity, thus potentially shifting or overcoming these "Role Limits ». In summary, "VUCA Role Limits" describe the point at which traditional or inadequately adapted systems in a complex, uncertain, and variable environment become overwhelmed because the challenges posed by the environment exceed their capacity for control and adaptation.

11. CPS (Cyber-Physical-Systems) Data Traits

1. Origin and Nature:

CPS data typically originates from sources such as sensors and IoT devices.

Common examples include Smart Cities or Connected Cars. This data is often referred to as *operational data*, characterized by the need for **real-time**, **high-quality**, and **reliable** processing. These expectations are particularly emphasized in the context of the **Model Context Protocol (MCP)**, which is used in industrial and IoT systems to facilitate seamless data exchange.

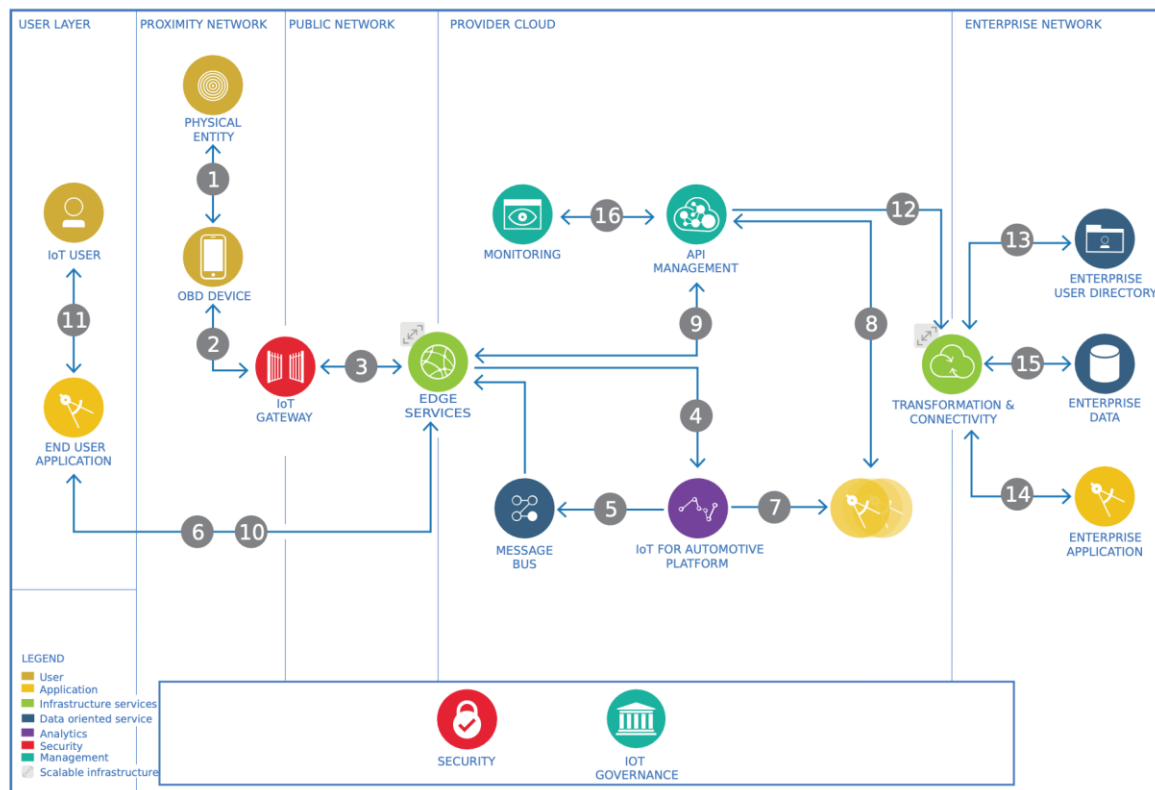


Fig: IoT Reference architecture IBM - Connected Cars

2. Fundamental Structure – Data DNA

Based on the “Data DNA” framework, which draws an analogy to biological DNA, the foundational building blocks of data in CPS are:

- **Entities** – Represent real-world objects or components (e.g., physical assets, sensors, control systems).
- **Properties** – Describe attributes of entities (e.g., a sensor's status, type, or location).
- **Relationships** – Capture interactions or associations between entities and their properties.

These components exhibit **heterogeneity** and **hierarchical coupling** across different levels. Each is subject to variations in type, structure, distribution, and organization, resulting in personalized traits for every instance. This structure supports domain-specific representations and promotes flexibility in CPS environments.

3. Timeliness and Processing:

CPS data must be processed and transmitted **quickly, securely, and reliably**. The **Model Context Protocol (MCP)** is designed to address this by enabling **interoperability**, low-latency processing, and **on-the-fly normalization** of streaming

data from various sources. It ensures seamless data transfer between systems such as industrial control units, enterprise IT platforms, and decentralized IoT devices.

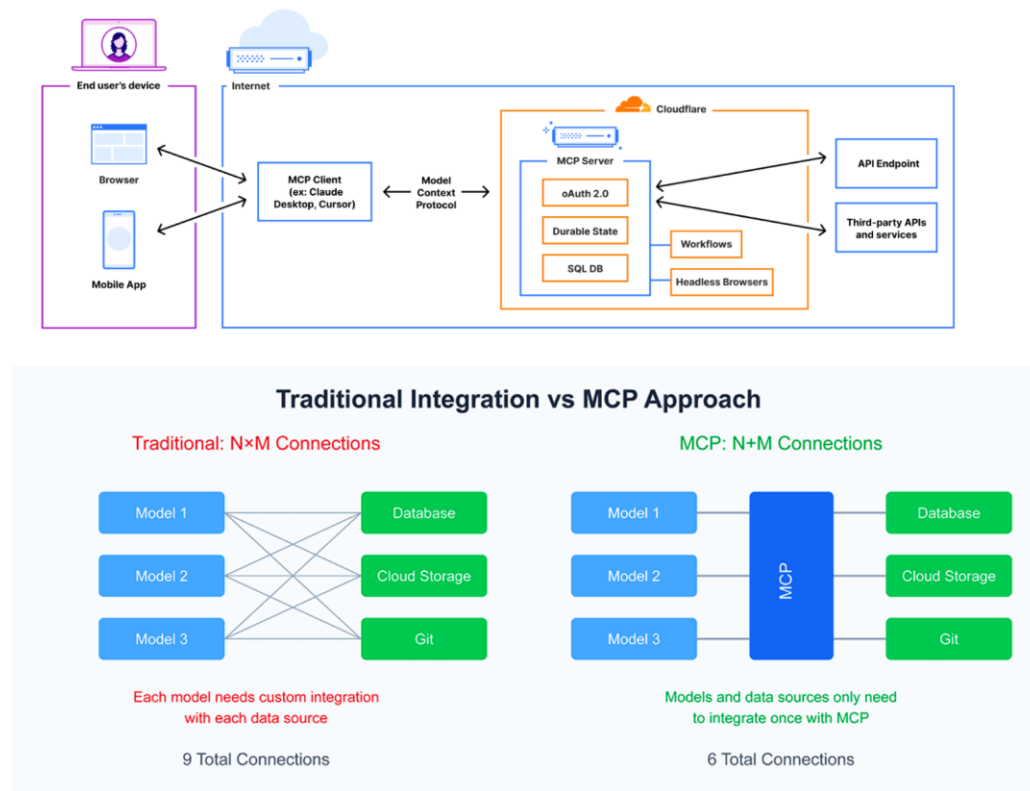


Fig: Model Context Protocol (MCP) Explained (Kelly, 2025)

4. Contextualization:

Understanding the **source**, **meaning**, and **metadata** of CPS data is essential for managing it effectively. MCP also plays a role in **contextualization**, embedding metadata to help clarify the purpose and lineage of data. As data flows scale in volume and complexity, strategies for managing metadata and importing it into metadata repositories—while balancing business and technical perspectives—become increasingly important.

5. Complexity:

CPS data contributes to the broader concept of **Data Complexity** and exhibits the following characteristics:

- **Large Scale & High Dimensionality** – Due to widespread sensor deployments.
- **Online & Real-Time Interaction** – Reflecting the dynamic nature of physical environments.
- **Mixed Sources** – Data streams originate from heterogeneous devices and platforms.

- **Strong Dynamics & High Frequency** – Continuous and rapid updates from systems.
- **Uncertainty & Noise** – Sensor errors and environmental variability introduce inaccuracies.
- **Unclear Structures & Hierarchies** – Especially in early-stage or diverse systems.
- **Strong Sparsity** – Incomplete observations or infrequent updates are common.
- **Heterogeneity** – Varied formats, structures, and semantics across devices.
- **Non-IIDness (Not Independently and Identically Distributed)** – Dependencies exist across time and context.

These properties impact the **value**, **quality**, and **interpretability** of data-driven processes in CPS environments.

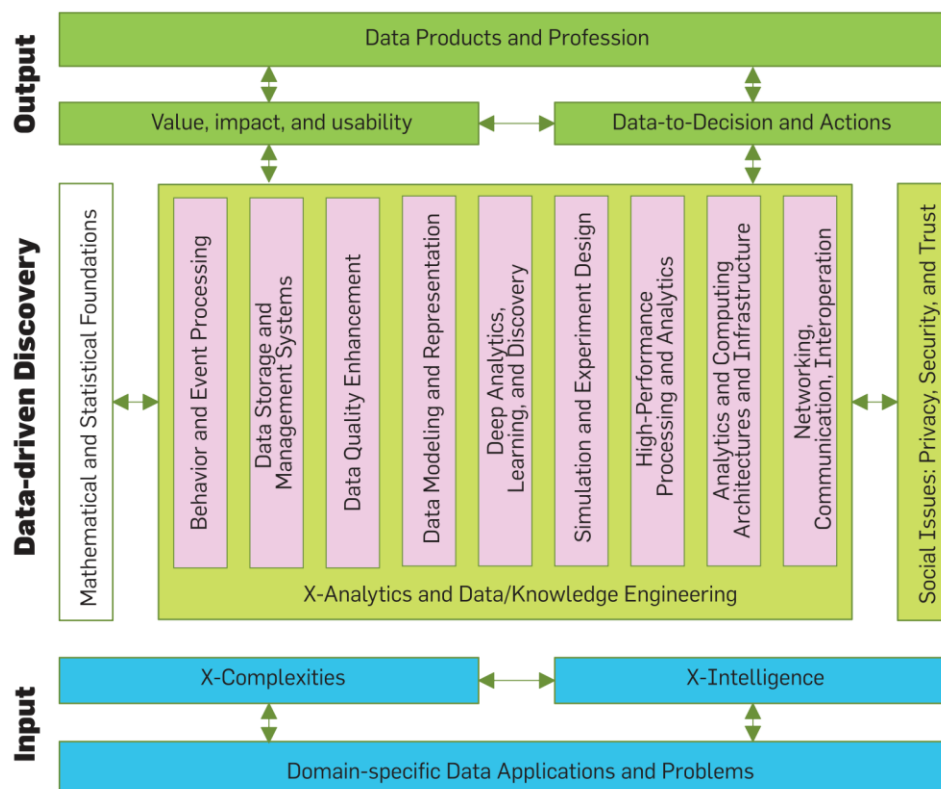


Fig: Data science conceptual landscape (Cao, 2017, 2018, pp. 63, 141)

6. Quality:

High **data quality** is essential for reliable analytics and decision-making in CPS. Quality is defined by how well the data meets user needs and use-case requirements. Key **Data Quality Dimensions** include:

- **Completeness** – Percentage of available data relative to expected volume.
- **Uniqueness** – Ensuring no duplication of records.
- **Timeliness** – Data reflects the current reality when needed. This includes:

- **Data Currency** – Are values up-to-date?
- **Latency** – Time between creation and availability.
- **Validity** – Adherence to defined formats, types, and ranges.
- **Accuracy** – Alignment with real-world measurements (impacted by noise or calibration).
- **Consistency** – Uniformity across different representations or data sources.

Data Integrity ties completeness, accuracy, and consistency together. Poor quality (e.g., missing values or outliers) can degrade CPS decision-making and fairness.

7. Processing and Analysis:

Managing CPS data involves various **preprocessing and analysis** techniques such as:

- **Data Cleansing, Parsing, and Profiling** – To handle data errors, anomalies, and format inconsistencies.
- **Pattern Detection, Trend Identification, and Anomaly Detection** – Especially relevant for applications like predictive maintenance or fault detection (e.g., in manufacturing systems).
- **System Dynamics Modeling** – A method used to understand CPS behavior by modeling causal relationships, feedback loops, and time delays using data from sources like numerical logs, simulation outputs, or mental models.

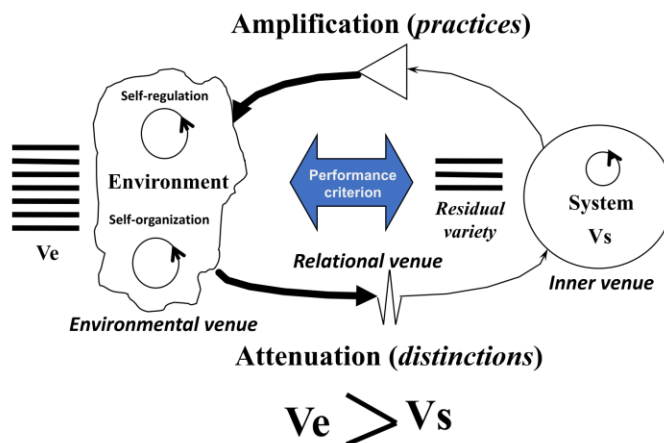


Fig: Control loop model as a template for management systems (Alvarado, 2022; Ashby, 1957; Lambertz, 2017, 1272, 202ff)

Summary:

Cyber-Physical Systems generate **dynamic, high-volume**, and often **complex operational data**. To extract value from such data for real-time monitoring, decision-making, and control, organizations must ensure:

- Fast and secure **processing**,
- Strong **metadata and context management**,
- Attention to **data quality and complexity**, and

- Application of suitable **analytical models and system dynamics tools**.

These traits make CPS data foundational for the development of intelligent, adaptive, and resilient systems in smart cities, industrial automation, and beyond.

12. Learning Types

Please check point 1.

[press here to go to point 1.](#)

13. Human-Centric Applications

Definition:

Human-centric applications refer to systems, tools, and approaches in data science and digital technologies designed to prioritize human needs, interaction, understanding, ethics, and usability. This theme was central in Week 12 of the course and emphasized in the exam as Topic 13.

1. Human Involvement in Data Science Systems

Human intelligence is fundamental to designing, interpreting, and using data-driven systems. This includes:

- **Explicit human input:** imagination, reasoning, supervision, evaluation, and expertise.
- **Implicit input:** emotional intelligence, intuition, collaborative ideation (e.g., brainstorming), and embodied cognition.

These ideas are foundational in **Human-in-the-Loop systems**, which combine algorithmic processes with human oversight to enhance trust, contextual understanding, and decision quality.

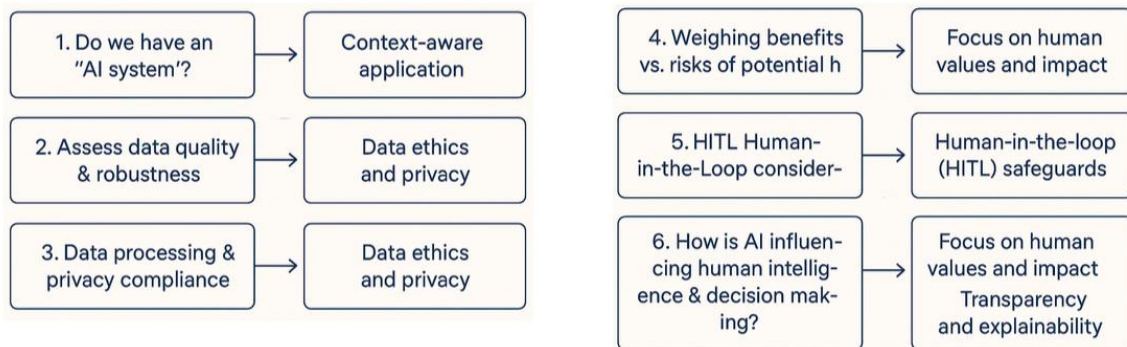
2. Human-Centric Design Frameworks

Week 12 introduced:

- **Design Thinking** as a method for problem-solving centered around user needs.
- A **six-step framework** for evaluating AI systems, addressing compliance, fairness, bias, and human oversight.

These are aligned with principles from:

- The **EU AI Act**,
- **GDPR** (privacy & data protection),
- **Ethical AI criteria** (e.g., the FAT framework: Fairness, Accountability, Transparency).



3. Interface, Usability & Access

Human-centric applications ensure that users can effectively interact with systems:

- **Dashboards** and visual tools combat data overload.
- **Role-based access** ensures the right people have the right data at the right time.
- **User experience design (UX)** tailors systems to skill levels and goals of various stakeholders.

4. Business Intelligence (BI) and Metadata

- Human-centric BI involves aligning tools (e.g., Tableau, Power BI) with user groups and decision-making needs.
- **Self-service BI portals** enable business users to perform analysis independently, promoting data democratization.
- **Metadata management** involves creating a shared **business glossary**, lineage visualizations, and intuitive tools to support communication across business and technical users.

5. Explainable AI (XAI)

AI explanations must be:

- **Interactive:** Allowing users to explore “why” and “how” a model made decisions.
- **User-specific:** Tailored to different knowledge levels, e.g., data scientists vs. decision-makers.

This is essential for **transparency, trust, and ethical deployment** of AI systems.

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PERFORMANCE PANEL	EVALUATION DETAILS <u>Evaluation Metrics</u> - details of evaluation metrics - information about test/validation set <u>Performance summary</u> - performance details on test/validation set - model performance visualization (expected failure and success) <u>Limitations</u> - limitations regarding the performance and possible error cases		
	DECISION EXPLANATION and EXPLORATION <u>Global Explanations</u> - representative cases - borderline cases - data manifold visualization <u>Instance Explanations</u> - instance level post-hoc explanations - comparison analysis with influencer samples from gallery set <u>Robustness Tests</u> - instance manipulation for sensitivity analysis - adversarial attacks - synthetic samples by deep generative models - decision review and correction for further model development		
MODEL PANEL	MODEL DETAILS <u>Model Info</u> - model type - model architecture - model visualization - model input and outputs - model behavior - model license <u>Development</u> - training framework - hyperparameters - hardware details - reproducibility and code reuse <u>Deployment</u> - framework - model input output details - mode size - inference time		

6. Communication & Change Management

Human-centric data systems also rely on:

- Clear, relatable communication of concepts like AI, data quality, and system goals.
- Adaptation to emotional and cognitive needs of users when introducing new tools or processes.

Conclusion

Human-centric design in data science ensures that **technology serves people**, not the other way around. It promotes ethical standards, intuitive use, and inclusivity across the entire data lifecycle — from collection to AI model outputs and decision-making.

Human-centered Design In A Nutshell

Human-centered design, also known as user-centered design, describes a framework that helps businesses understand the needs, wants, contexts, constraints, and behaviors of their users. Human-centered design (HCD) is an approach that makes users the priority to develop useful products and services.

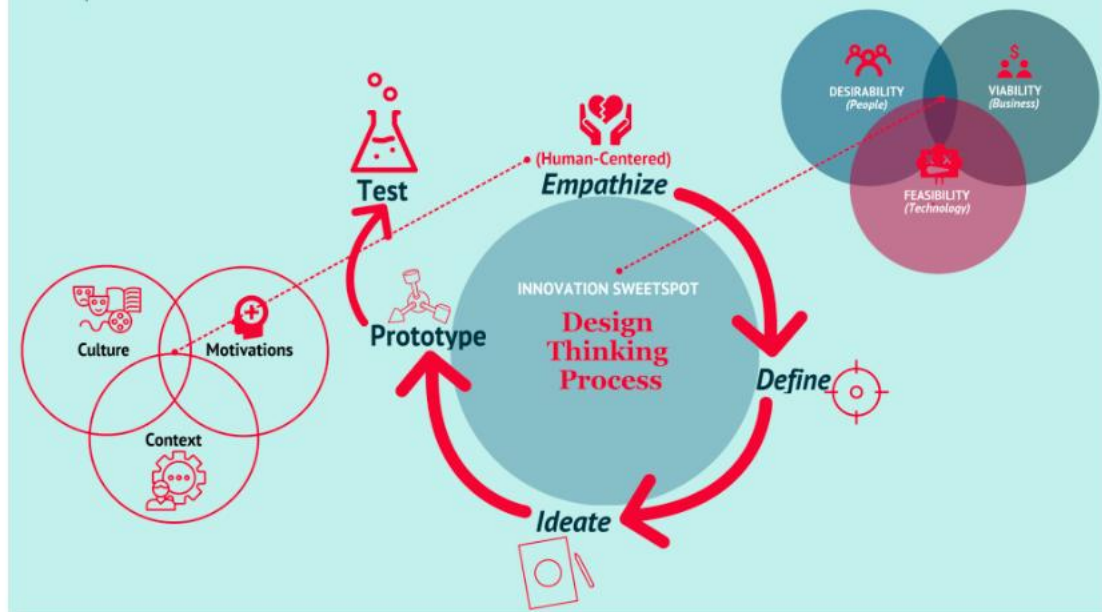


Fig: Human Centric Design Framework ("IDEO Design Thinking | IDEO | Design Thinking", n.d., online)

14. VUCA Challenges

Definition:

VUCA stands for **Volatility, Uncertainty, Complexity, and Ambiguity** — a framework used to describe challenging conditions in environments where decision-making is difficult due to instability, unpredictability, and interdependencies.

1. Context and Relevance

- Covered in **Week 3** under "VUCA and the Necessity of Data Science".
- Listed as **Topic 14** in the exam overview.
- Closely tied to themes like **decision quality, resilience, and system dynamics** in complex environments.

2. Characteristics of VUCA Dimensions

Category	Description
Volatility	Unstable conditions, but not necessarily complex (e.g., sudden supply chain disruptions).

Category	Description
Uncertainty	Future outcomes are unknown despite available data (e.g., competitor reactions).
Complexity	Many interrelated variables; cause and effect are not obvious (e.g., global regulatory compliance).
Ambiguity	Lack of clarity due to unknown unknowns; no precedents (e.g., launching a radically new product).

A Guide to Approaching Events in the Four VUCA Categories



Fig: Understanding threats to performance in a VUCA world (ad. Bennett & Lemoine, 2014, p. 3)

3. Why Data Science is Needed in VUCA Conditions

- Traditional decision-making often fails under VUCA. Data science provides tools to **sense**, **model**, and **respond** dynamically.

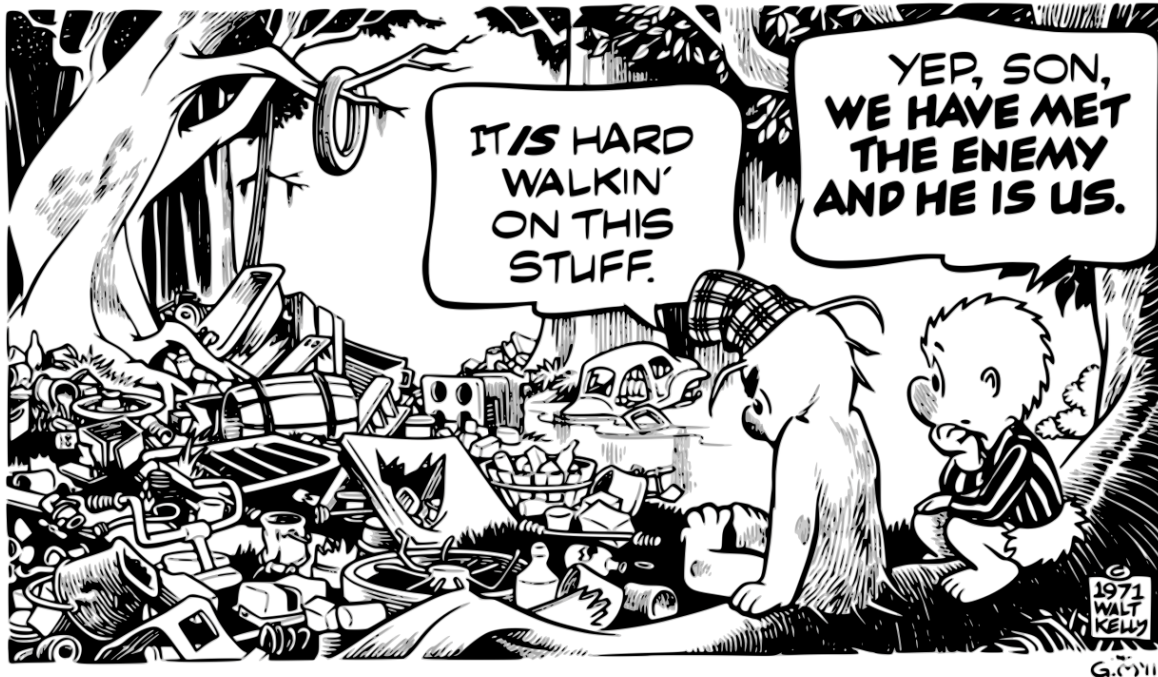
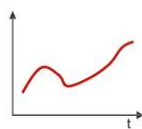


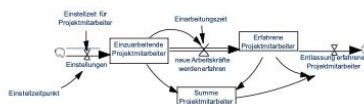
Fig: System Dynamics is the “causal” view of the world. ad. Fig: “Pogo” by Walt Kelly (Earth Day 1971)

- Enables **evidence-based action** in the face of limited predictability and high stakes.
- **System dynamics** help visualize feedback loops, delays, and interactions in complex systems.

P'roblem
("reference mode")

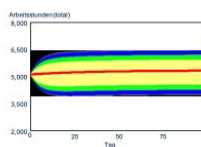


Hypotheses
(= the formal model)

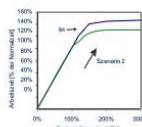


Analyses

(simulation experiments)



Policy design
(testing improved structures)



Implementation



Fig: PHAPI - System dynamics process (J. W. Forrester, 1994, p. 245)

4. Key Course Questions (Week 3)

- How does system dynamics assist in navigating VUCA?
- What are the modern drivers of VUCA (e.g., globalization, digital disruption, climate uncertainty)?
- How can organizations build **resilience** using data-driven approaches?
- What role does **interdisciplinary thinking** play in managing complexity?

15. Ethical Cooncern

Definition and Importance

Ethics in data science involve principles that guide responsible behavior, rooted in fairness, respect, integrity, transparency, and accountability. Ethical data handling encompasses the full data lifecycle—from **collection and storage** to **analysis, usage, and disposal**—and ensures alignment with societal values and stakeholder trust.

Unethical practices not only pose legal risks but can harm an organization's reputation and undermine public confidence. Ethics in data science are not optional; they are a **core element of social responsibility**.

1. Why Ethics Matter in Data Handling

- **People are impacted:** Data often represents real individuals and drives decisions affecting their lives.
- **Misuse is risky:** Data can easily be misrepresented, manipulated, or used in harmful ways.
- **Data has economic value:** Raises questions about ownership, access, and fairness in value distribution.

Handling data ethically builds long-term organizational resilience and trust.

2. Privacy and Regulatory Foundations

Privacy is a **fundamental human right**. Ethical data handling includes:

- Protecting **personally identifiable and sensitive information**
- Complying with laws such as:
 - **GDPR (EU)**
 - **US Privacy Act (1974)**
 - **HIPAA, FERPA, SOX, GLBA** (sector-specific)
 - **OECD's Eight Principles:** collection limits, security, accuracy, purpose limitation, and accountability

Regulations help balance **individual protection** with **legitimate data usage**.

Applies to AI and Data Science:

- Inclusive growth and well-being
- Human-centered values
- Transparency and explainability
- Robustness and safety
- Accountability

3. Ethical Frameworks for Data Science & AI

- **Traditional Ethical Theories**
 - *Utilitarianism*: Maximize benefit (e.g., for public good)
 - *Deontology*: Respect for rules/rights (e.g., user privacy)
 - *Virtue Ethics*: Focus on the scientist's integrity and honesty
- **FAT Framework (Zhdanov et al., 2022)**
 - **Fairness**: Avoid bias
 - **Accountability**: Who is responsible when things go wrong?
 - **Transparency**: Make decisions understandable

Focuses on:

- **Fairness** – Avoid bias and ensure equitable treatment.
- **Accountability** – Who is responsible when algorithms go wrong?
- **Transparency** – Make models and data decisions understandable.

XAI presents a set of techniques and methods to understand why AI makes a particular decision. Knowledge of the reasons for a decision is considered key to understanding model decisions and improving users' trust in AI-based solutions.

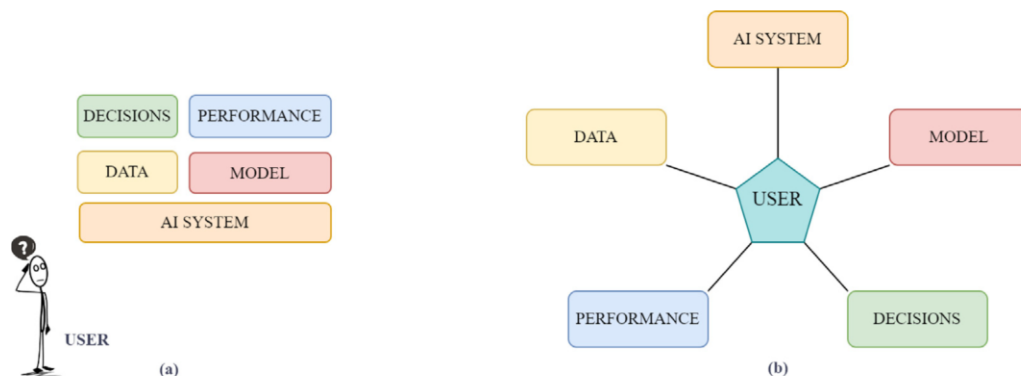


Fig: Design of two explainability approaches in AI: (a) a conventional approach that does not consider the different user groups when presenting explanations, and (b) a user-centric approach that provides explanations tailored to the requirements of different user groups. (Kaplan et al., 2024, p. 5)

4. Key Ethical Risks in Practice

- **Misrepresentation:** Cherry-picking data or misleading visualizations
- **Bias:** Reinforcing societal prejudice under a “neutral” guise
- **Statistical misuse:** Presenting weak correlations as causation
- **Lineage & integration risks:** Altered or undocumented data origin
- **Anonymization failures:** Re-identifying individuals in large datasets
- **Big Data risks:** Over-surveillance, skewed samples, opaque filters
- **LLM (e.g. GPT) risks:** “Stochastic parrots” spreading misinformation
- **Societal consequences:** Job displacement, increased inequality

These risks require both technical safeguards and ethical reflection.

5. Building an Ethical Data Culture

To embed ethics in practice:

- Define expected behaviors in a **code of conduct**
- Conduct **training** for analysts, data scientists, and leadership
- Implement **whistleblower protection**
- Align with **Organizational Change Management (OCM)** principles

Data governance plays a critical role:

- Establish standards, review decisions, enforce oversight
- Incorporate access controls, auditing, and accountability measures

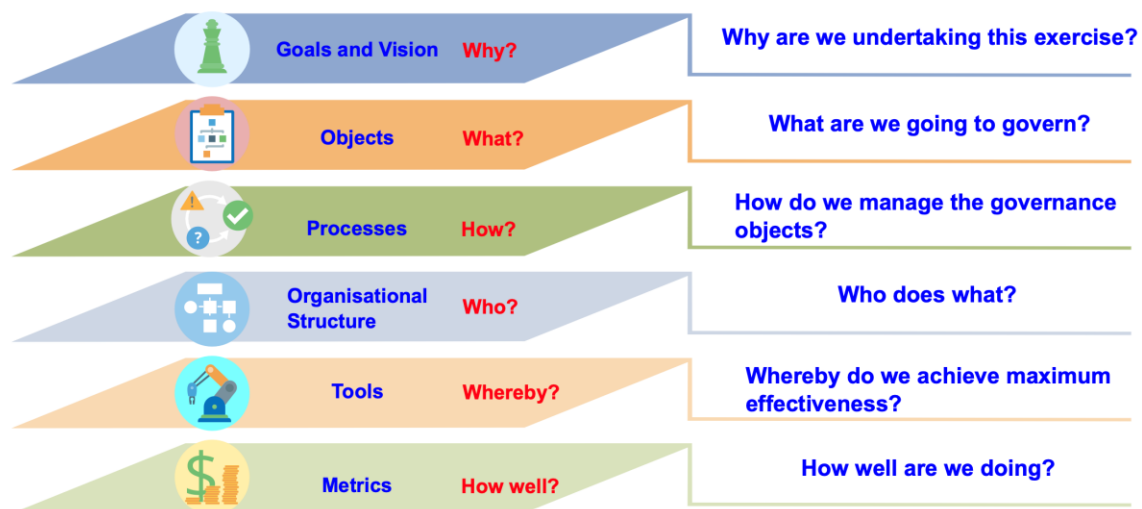


Fig: Conceptual Data Governance model (International et al., 2017)

Conclusion

Ethics is not a checklist—it is a **foundational principle** that underpins **trustworthy, responsible, and sustainable** data science. From fairness and transparency to privacy and accountability, ethical awareness must guide every step of the data journey.

16. CPS (Cyber-Physical-Systems) Feedback

Definition and Relevance

Cyber-Physical Systems (CPS) are integrations of physical components (like machines, sensors, actuators) with cyber capabilities (data processing, AI, networking). Feedback mechanisms are central to CPS design: they **enable the system to adjust based on real-time data about its own state or performance**.

The term **xCPS Feedback**, as used in this course, likely refers to **extended or cross-domain feedback processes** within CPS — especially in data-rich, real-time environments.

1. Feedback in CPS: The Foundation of Control

- **Feedback loops** are mechanisms that monitor outputs and feed them back as inputs to adjust system behavior.
- They are essential for **system stability, adaptation, error correction, and automated decision-making**.

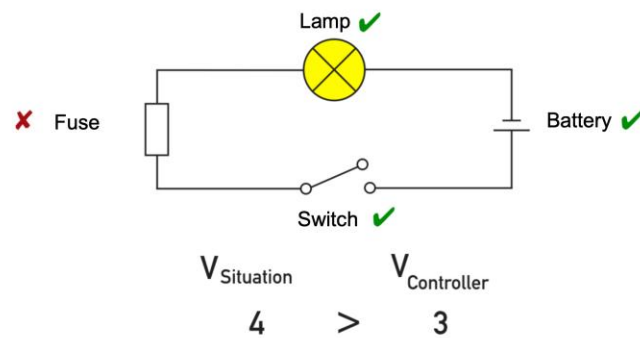
- In CPS, feedback connects:
 - **Sensors** (collect data from environment)
 - **Controllers** (analyze data and decide actions)
 - **Actuators** (execute decisions in the physical world)



Example: A smart thermostat adjusts heating based on real-time temperature feedback from sensors.

2. System Dynamics and Control Models

- **System dynamics** is the study of how systems evolve over time using stocks, flows, and feedback loops.
- The **Control Loop Model** is often used in CPS to describe how inputs, controllers, and outputs interact through continuous feedback.
- These models help design, simulate, and test CPS behavior.



In the last example, a hidden component comes into play: In this scenario, there is still a fuse inside the circuit - only unfortunately, you did not have a mental model of the situation that provides for such an element. Thus, you are denied control over the situation ("Repair circuit!"). The variety of the situation is greater than the variety of the controller.

This illustrates a breakdown in control due to incomplete system knowledge — a feedback design flaw.

3. The Viable System Model (VSM)

- A **recursive management structure** for complex systems
- Incorporates **anti-oscillatory feedback loops** to ensure adaptability, learning, and system health
- Maps well to **multi-level CPS** where different subsystems monitor and regulate each other.

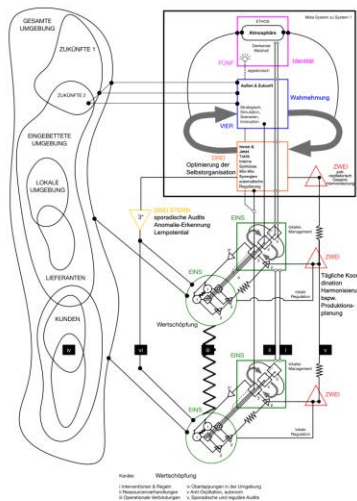
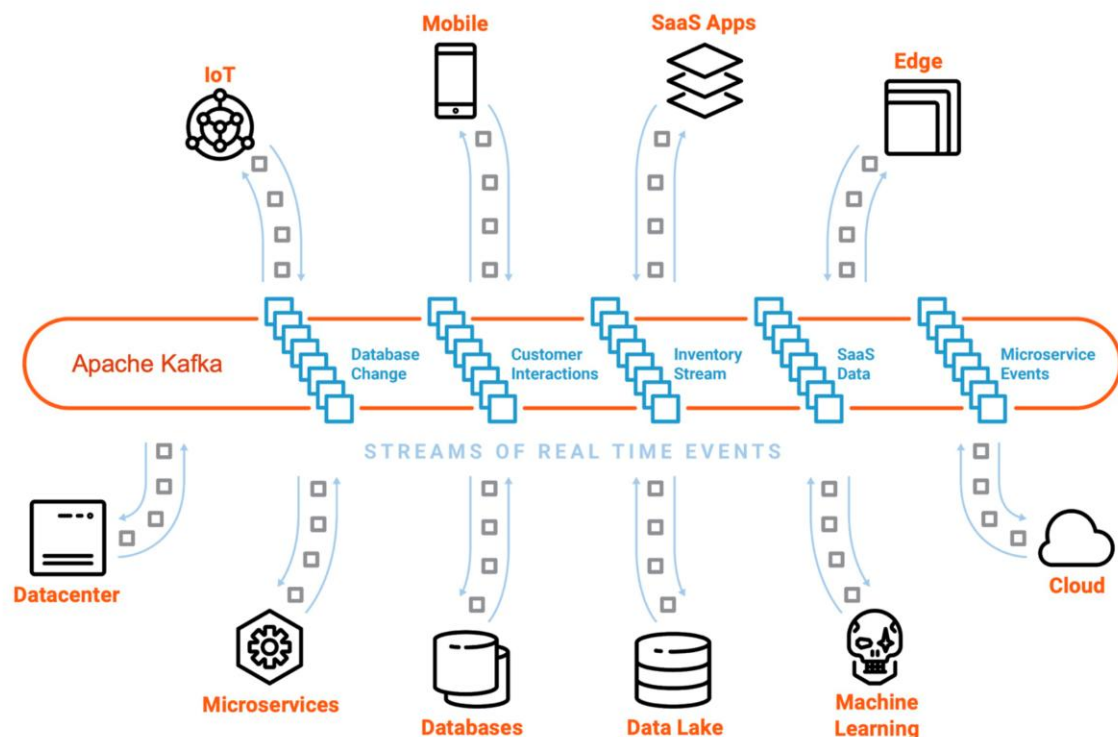


Fig: **Recursive** Viable System Model (VSM) (Beer, 1981, p. 130; Schwaninger, 2018, p. 99; Funk and Schmidt, 2020, online; Medina, 2011; Medina, 2015, online; Beer, 1972, online)

4. Event Streams & Real-Time Feedback

- CPS generate data in the form of **event streams** (continuous signals, sensor data, status updates)
- Feedback in these systems is:
 - **Fast** (milliseconds to seconds)
 - **Continuous** (always on)
 - **Data-rich** (multi-sensor integration)
- **Data Science tools** (stream processing, anomaly detection, ML-based control) are key to making this feedback actionable.



Conclusion

CPS feedback is what makes **automated, adaptive control possible**. Whether modeled using system dynamics, implemented in control loops, or supported through data stream analytics, **feedback is the invisible engine** that allows CPS to operate intelligently, safely, and autonomously.

17. Modeling Types

1. What is a Data Model?

A **data model** is a structured representation of data, often serving as a **blueprint** for how data is collected, stored, related, and used.

Data modeling is the process of discovering, analyzing, and formalizing these data requirements—critical to communication between business and technical teams.

Purpose: Provide a shared vocabulary, document business rules, and improve reusability, governance, and integration.

2. Core Components of Data Models

- **Entity**: An object or concept of interest (e.g., Customer, Product)
 - **Attribute**: Properties or fields describing entities
 - **Relationship**: Associations between entities
 - **Domain**: Allowed values or rules for attributes
-

3. Levels of Data Models

Model Type	Purpose
Conceptual	High-level overview, representing real-world entities and relationships
Logical	Technology-independent detail: attributes, keys, normalization
Physical	Technology-specific schema (e.g., tables, columns)
Canonical	Generic transport model for system interoperability
Views	Virtualized representations for querying or access control

Each layer moves from abstract (business) to concrete (technical implementation).

4. Modeling Schemes and Notations

- **Relational**: IE notation (e.g., crow's feet); used in RDBMS
 - **Dimensional**: Star/Snowflake schemas (BI/data warehousing)
 - **Object-Oriented**: UML class diagrams
 - **Fact-Based**: ORM, FCO-IM (natural language-driven)
 - **Time-Based**: Anchor Modeling, Data Vault (temporal tracking)
 - **NoSQL**: Graphs, documents, key-value, columnar
-

5. Extended Model Types & Architectures

- **Data Architecture Models:** Enterprise-wide blueprints (e.g., Zachman Framework, EDM)
 - **Reference & Master Data Models:** Govern consistency and canonical structure for shared domains
 - **Governance/Operating Models:** Centralized, decentralized, federated structures
 - **Metadata & Content Models:** Ontologies, taxonomies, thesauri for semantic layer
 - **Quality Models:** Dimensions (completeness, accuracy), Model Scorecards
 - **Data Flow Models:** Show movement across systems, pipelines, APIs
-

6. Data Science & AI Models

- **Analytic Models:** Clustering, association, self-organizing maps
 - **Evaluation Tools:** Confusion Matrix, ROC curves
 - **Learning Types:** Supervised, Unsupervised, Reinforcement, Zero-shot
 - **Agents:** Goal-based agents as dynamic modeling units
 - **Feature Types & Leakage:** Crucial for valid modeling pipelines
-

7. System Dynamics Models

- Use **feedback structures** (e.g., balancing vs. reinforcing loops)
 - View the system holistically using:
 - Mental, written, or numerical models
 - Control loop templates (monitor → evaluate → act)
-

8. Tools & Standards

- **Modeling tools:** ER/Studio, PowerDesigner, Sparx EA
- **Standards:** Naming conventions, normalization rules, review checklists
- **Life cycle:** Plan → Build → Review → Manage

Limitations: Models require maintenance, don't capture every scenario, and may be hard to interpret without context.

18. System Thinkings

1. What Is Systems Thinking?

Systems Thinking is a **causal and holistic approach** to understanding complex systems by focusing on **interrelationships and patterns**, not just parts. It integrates:

- **Reductionism** (breaking things down)
- **Holism** (understanding the whole)
→ via a methodology called **systematism**.

Goal: Understand why systems behave the way they do — not just predict what will happen.

2. Key Concepts from System Dynamics

System Dynamics is a simulation-based methodology combining **qualitative insights** with **quantitative modeling**. It uses:

- **Feedback Loops:**
 - Balancing (negative): “The more A, the less B”
 - Reinforcing (positive): “The more A, the more B”

Often modeled in the **Recursive Viable System Model** or **Control Loop Models**

- **Delays:**
Time lags that change system behavior
- **Stocks and Flows:**
Stocks = accumulations (e.g., inventory)
Flows = rates of change (e.g., sales per day)
- **Nonlinearity:**
Real-world cause-effect is rarely linear.

Tools: **Causal Loop Diagrams, Stock-Flow Diagrams**, modeled in software like **Vensim**

3. Mental Models

Mental models are internal belief systems or assumptions that influence how people interpret and act within systems.

Types of responses to system situations:

- **Reactive** (event-based)
- **Adaptive** (pattern-based)
- **Creative** (structure-focused)
- **Generative** (change-worldview-focused)

Shared mental models support organizational learning.

4. Applications of Systems Thinking

Systems Thinking is used to analyze:

- **Supply Chains** (Bullwhip Effect)
- **Traffic Flow and Congestion**
- **Sustainable Resource Management**
- **Organizational Adaptation in VUCA conditions**

Especially useful for:

- Diagnosing unintended consequences
 - Designing **long-term strategies** over short-term fixes
-

5. Organizational Design and Architecture

- Used to **map connections between data, business processes, and technology**
 - Supports **enterprise-level strategy through systems design**
 - The **St. Gallen Management Model** is a well-known framework connecting environment, stakeholders, and structure.
-

6. Systems in Computation

- **Multi-Agent Systems**: Software agents acting autonomously in complex environments
- **Production Systems**: Holistic, socio-technical models requiring structural flexibility

19. Transparency & Trust

1. Transparency: Definition and Importance

Transparency refers to the openness, visibility, and understandability of information and practices. It is a foundational principle in ethical data handling, data science, and organizational communication.

Key areas of transparency:

- **Data Ethics and Privacy:**
Organizations must clearly communicate how personal data is collected, used, and protected. Transparency supports individual dignity, autonomy, and informed consent.
 - **Data Warehousing & Business Intelligence (DW/BI):**
Providing metadata, usage context, and access to data lineage improves confidence. Dashboards and monitoring tools make internal processes visible, supporting stakeholder trust.
 - **AI and Data Science:**
Transparency includes:
 - **Explainability:** Users must understand how AI makes decisions.
 - **Traceability:** System logic and outcomes must be accountable and auditable.
 - **OECD and UN principles** emphasize explainability at all AI lifecycle stages.
 - **Communication Theory:**
Effective communication is not only about what is said but what is understood. For transparency to be meaningful, information must be communicated in a way people can correctly interpret.
-

2. Trust: Definition and Role

Trust is the belief in the reliability and integrity of data, systems, and people. It is both a technical and social construct, built on:

- **Data Quality:** Trust begins with accurate, consistent, and relevant data.
- **Ethical Practices:** Respecting privacy, avoiding misuse, and adhering to regulations helps preserve public trust.
- **Organizational Culture:** Trust within teams is built through honesty, shared goals, and transparent decision-making.
- **Change Management:** Teams are more effective when trust exists between leadership and employees.
- **AI Systems:** Explainability directly impacts users' trust in algorithmic decisions.
- **Communication:** Miscommunication weakens trust; shared understanding strengthens it.

A foundational concept is the “**True and Fair View**”: information must be factually accurate (true) and presented in a way that can be interpreted appropriately (fair).

3. The Link Between Transparency and Trust

Transparency and trust are deeply interdependent:

- **Transparency fosters trust:** Openness about data practices, system logic, and decisions builds user and stakeholder confidence.
- **Trust requires communication:** The quality and interpretation of information determine how much people trust systems and one another.

In ethical AI, transparency (e.g., explainable models) is essential for trustworthiness.

20. Data Bias

1. Definition and Significance

Data bias refers to any systematic error in data collection, selection, analysis, or presentation that leads to **distorted, misleading, or unfair outcomes**. It compromises objectivity and reduces the validity of decisions or insights derived from data.

Bias may arise from:

- Poor sampling methods
- Human assumptions and heuristics
- Algorithmic learning from skewed data
- Intentional manipulation of data for a narrative

It can affect **scientific integrity, public trust, fairness, and AI ethics**.

2. Common Sources and Types of Bias

Type of Bias	Description
Reporting Bias	Overrepresentation of extreme or memorable events
Historic Bias	Inherited societal or cultural inequities embedded in old data
Confirmation Bias	Analysts search for or emphasize data that supports their preconceptions
Observer Bias	Researchers' expectations influence collection or interpretation
Sampling Bias	Sample is not representative of the target population
Survivorship Bias	Only considering "successes" while ignoring failures
Hunch & Search	Selective data usage driven by gut feeling or expected outcome
Data Processing Bias	Data transformation/cleaning skews results
Algorithmic Bias	Models reproduce and amplify biases in training data
Intentional Collection Bias	Data is gathered or used to reinforce a pre-defined result

Bias often begins with **human influence** but becomes more dangerous when **automated and scaled by AI**.

3. Risks and Impact

- Misrepresentation of reality
- Reinforcement of discrimination or historical injustice
- Erosion of public trust
- False appearance of neutrality
- Ethical violations in high-stakes use cases (e.g., criminal justice, hiring, credit scoring)

Unchecked bias reduces **transparency**, harms vulnerable populations, and leads to poor or unethical decisions.

4. Addressing Bias: Ethical and Practical Imperatives

Ethical Foundations:

- Justice and fairness require identifying and mitigating bias.
- Especially critical for **high-risk applications** (e.g., predictive policing, loan decisions).

Transparency:

- Bias mitigation starts with acknowledging uncertainty and documenting assumptions.
- Ensuring explainability of models and clarity in data processing builds **algorithmic accountability**.

Data Quality Assessment:

- Spotting and correcting bias is a fundamental step in data validation and ethical analysis.
-

5. Summary

Bias is not only a technical flaw but a **social, ethical, and organizational risk**. Responsible data science demands:

- Rigorous design
- Transparent methodology
- Ethical awareness
- Ongoing monitoring of models and inputs

A **fair and trustworthy data ecosystem** depends on understanding and actively combating bias.

21. Importance and Methods of Problem Definition in Data Science and Problem-Solving

Defining the problem is a crucial first step in data science and general problem-solving. A well-defined problem guides model selection, data collection, and overall project direction, while a poorly defined one risks failure and misallocated resources.

Nature of Problems

Problems can range from simple "difficulties" with clear solutions to complex "messes" with interrelated factors and human interests. Data science problems often fall into the latter, involving system-level complexities (X-complexities) and requiring systemic, not linear, thinking. Frameworks like VUCA (Volatility, Uncertainty, Complexity, Ambiguity) further stress the importance of tailored problem understanding.

Challenges

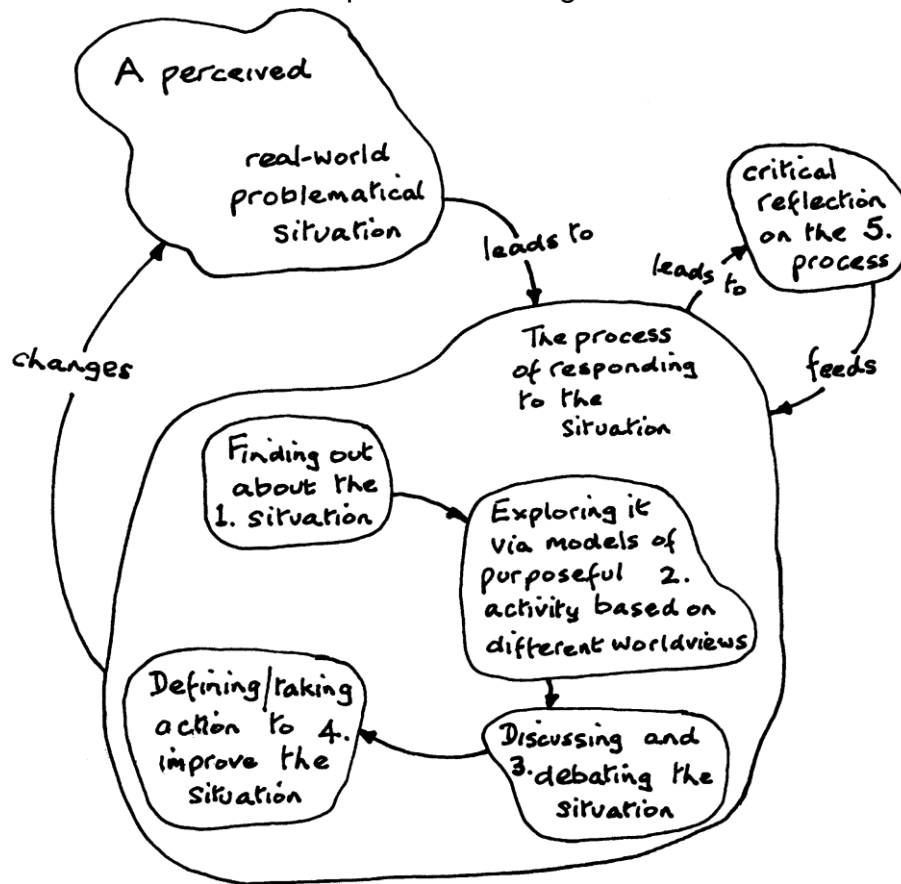
- Defining problems is difficult due to:
- Conflicting stakeholder views
- Ambiguous language
- Resistance to change
- Incomplete knowledge and differing values

Problem Definition Methods

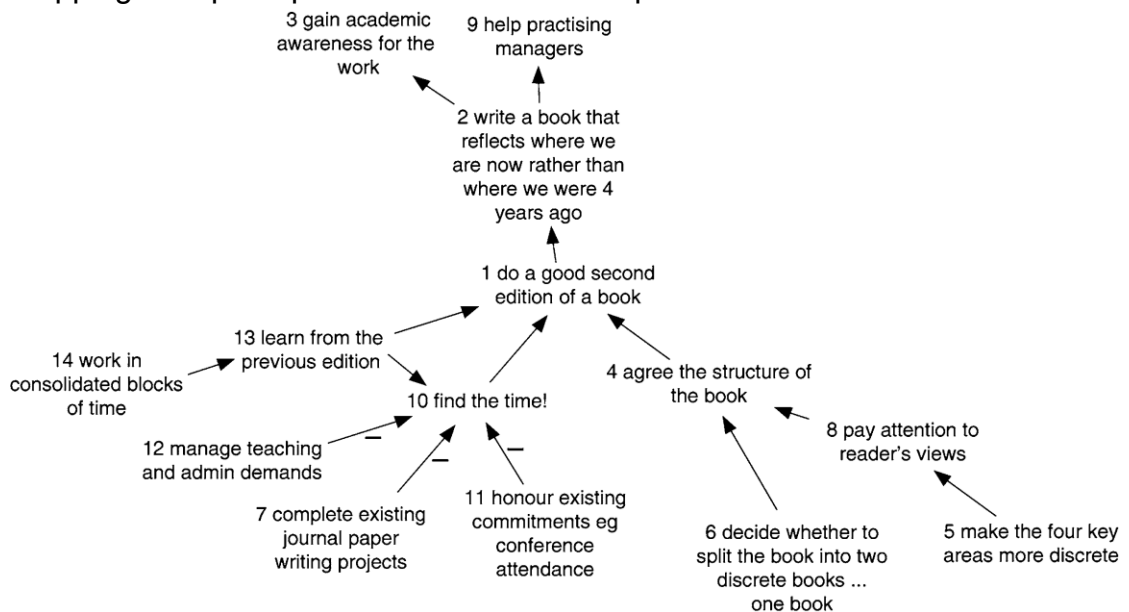
Several structured approaches exist:

- General Steps: Include identifying stakeholders, understanding system boundaries, clarifying objectives, and separating real from pseudo-problems.
- System Dynamics: Focuses on conceptual models, dynamic hypotheses, and causal loop diagrams.
- Soft Systems Methodology (SSM): Emphasizes capturing multiple perspectives using tools like Rich Pictures and iterative redefinitions.

- Critical Systems Heuristics (CSH): Uses boundary critique to question what's included or excluded in problem framing.



- Strategic Options Development and Analysis (SODA): Applies causal mapping to explore problem structures in depth.



Problem Definition in Data Science

In data science, problem definition should be problem- and goal-driven. It involves analyzing gaps between problem complexity and current capabilities, and iteratively

refining understanding through data and feedback. Rather than seeking a single solution, the goal is often to improve the situation through informed action.

22. Common Misconceptions About VUCA

Many leaders misunderstand VUCA by treating Volatility, Uncertainty, Complexity, and Ambiguity as interchangeable or as a single vague concept. This oversimplification leads to several key issues:

- **Lumping VUCA Components Together:** Treating the four distinct elements as synonyms undermines the ability to diagnose and respond effectively to challenges.
- **Belief That Preparation Is Futile:** Some assume that VUCA conditions make planning pointless, which discourages strategic thinking and readiness.
- **Generic, One-Size-Fits-All Solutions:** Responses like "be creative" or "be flexible" are too broad and ineffective across the distinct VUCA domains.
- **Misdirected Resources:** Misdiagnosing the specific type of VUCA situation (e.g., confusing complexity with uncertainty) leads to poor strategic decisions and wasted resources.

Core Insight: Each VUCA element requires a unique approach, and failing to differentiate them leads to confusion and ineffective action. Accurate diagnosis is essential for targeted, successful responses.

23. Myths and Misconceptions About Big Data and Data Science

The terms "Big Data" and "Data Science" have become popular buzzwords, often surrounded by hype, misunderstanding, and overuse. This has led to confusion, inflated expectations, and a lack of clear consensus on their actual meaning and purpose. Many common myths and misconceptions have emerged, which obscure the real challenges and opportunities these fields offer.

Key Misconceptions About Big Data

Big Data is Only About Volume

- While volume is a well-known feature, data complexity is often more defining than size alone.
- A dataset qualifies as "big" not just when it is large, but when its size becomes central to the problem being solved.

Big Data Equals Data Science or Analytics

- These terms are often wrongly used interchangeably.
- Big Data relates to data characteristics and infrastructure, while Data Science is a broader discipline involving methods, theory, and application.
- Confusing them oversimplifies their respective roles.

Big Infrastructure is Required

- It is not essential to have huge datasets or high-performance infrastructure to conduct meaningful Big Data research.
- Theoretical or methodological work, as well as many practical applications, can be done on smaller scales.

More Data is Always Better

- Bigger is not always better. The relevance and quality of data matter more than sheer volume.
- Effective data use requires targeted collection and smart handling, not just accumulation.

Data Quality is Unimportant

- Some assume that volume compensates for poor data quality, but this is false.
- Veracity (data reliability and trustworthiness) is one of the essential dimensions of Big Data.
- High-volume, low-quality data can lead to unreliable or harmful outcomes.

Popular Myths Summarized

- Size is all that matters.
- The main challenge is technical (architecture, algorithms).
- Analytics is the core issue.
- Data reuse is easy and obvious.
- Data Science = Big Data.
- Big Data is all marketing hype.

Focus on Structured, Predictive Analytics

- Marketing often highlights structured data and predictive analytics, but this ignores that ~95% of big data is unstructured (e.g., text, audio, images).
- Overemphasizing structured data distorts the scope of Big Data applications.

Single Version of the Truth (SSOT) Assumption

- It's a myth that merging data sources will naturally yield a complete and accurate "Golden Record".
- Multiple Versions of the Truth (MVOT) may be more realistic, especially when dealing with complex, fragmented data environments.

Data Lakes Are Inherently Valuable

- Without proper metadata management, data lakes can devolve into data swamps, making it difficult to retrieve and use the stored data effectively.
- Poor integration and lack of standards create usability challenges.

Consequences of Misuse and Overhype

- These misconceptions have led to an overuse and misuse of the terms "Big Data" and "Data Science," which clouds meaningful progress.

- Public and academic discourse often fails to reflect the true complexity and scope of these fields.
- Google search results, for example, often surface irrelevant or superficial content, further distorting public understanding.

Related Pitfalls

- Spurious correlations in large datasets can lead to false conclusions if not handled correctly.
- Bias embedded in large, uncured datasets can result in unfair or harmful outcomes, especially in sensitive domains like healthcare, criminal justice, and hiring.

Conclusion

Clarifying the definitions, boundaries, and practical implications of Big Data and Data Science is crucial for the field's healthy development. Dispelling these myths allows researchers, businesses, and policymakers to make informed decisions, focus on meaningful innovation, and avoid wasted efforts or harmful outcomes caused by blind reliance on buzzwords and hype.

24. Model Evaluation Pitfalls

1. Fundamentals of Model Evaluation

The quality of data-driven results is largely determined by how well a model aligns with the ground truth. Evaluation involves structured testing and expert analysis.

- Model performance is measured using metrics such as accuracy.
- CRISP-DM defines evaluation as the testing phase aligned with a predefined design.
- Includes behavioral interpretation and assessment of model outcomes.
- Expert feedback and deviations from the test plan are also taken into account.

2. Gaps in the Classical Data-Driven Approach

Traditional approaches reveal significant limitations regarding model reliability, trust, and adaptability to real-world complexities.

- Imbalance between models, outcomes, and decision-making.
- Low dependability (consistency over time and contexts).
- Poor repeatability (inability to reproduce results).
- Limited trust in model results.
- Lack of explainability for end users and stakeholders.
- Weak deliverability and transferability to other use cases.
- Inability to fully capture complex problem domains and data intricacies.

3. Interpretation Challenges in Complex Models

Modern models, especially foundation models, are hard to interpret due to their size and complexity. Human-centered explanation remains a challenge.

- Large parameter spaces and non-transparent interactions hinder understanding.
- Difficult to trace the logic behind specific predictions or outcomes.
- Explainable AI (XAI) often focuses on technical factors (e.g., scalability, output clarity).
- Human cognitive processes and true interpretability are frequently overlooked.
- There is a growing need for robust, user-focused evaluation metrics.

4. Pitfalls in Visualizing Model Outputs

Visualizations, while helpful, can distort interpretation if poorly designed or misused.

- May manipulate emotions rather than inform objectively.
- Risk of ambiguity and misinterpretation between sender and receiver.
- Require visual literacy; interpretation is not always intuitive.
- Can imply false completeness or logic.
- Vulnerable to unconscious or deliberate bias in design or presentation.

5. Modeling Process Pitfalls

Errors in the modeling approach itself can undermine evaluation, especially when real-world systems are mirrored uncritically.

- Common mistake: modeling real systems directly instead of conceptual abstractions.
- Leads to replication of existing conditions without critical assessment.
- Limits the model's ability to challenge current practices or reveal new insights.

6. Data-Related Pitfalls

Faulty or outdated data can silently undermine model evaluation by distorting both input and test outcomes.

- Errors such as duplicate entries or untimely updates often go unnoticed.
- These data issues feed directly into training or evaluation, skewing results.
- The impact on model performance may not be immediately visible but remains critical.

25. CPS (Cyber-Physical-Systems) Control

What is a CPS ?

A **Cyber-Physical System (CPS)** is a system that tightly integrates physical processes (machines, sensors, people, environments) with cyber components (data processing, control systems, communication technologies). CPS enable real-time monitoring, analysis, and control of the physical world through digital means.

- **MES (Manufacturing Execution Systems)** are a typical CPS application.
- Link production machinery (physical) with ERP systems (cyber).
- Enable real-time monitoring, control, and optimization of production.

Core Components:

- **Physical layer:** Machines, devices, sensors, actuators.
- **Cyber layer:** IT systems, data analytics, control algorithms.
- **Interfaces:** Bidirectional data flow between physical and digital layers.

Key Concepts:

- **Feedback loops:** Continuous input from the physical world informs system behavior.
- **Model-based control:** Complex systems require accurate models to remain manageable.
- **Cybernetics & Systems theory:** Foundations for controlling dynamic and complex systems.
- **Event/Data processing:** Techniques like CEP (Complex Event Processing) and SPC (Statistical Process Control) ensure responsive and predictive control.

Challenges:

- Managing **complexity** and system **variety** (Ashby's Law of Requisite Variety).
- Achieving **integration** of heterogeneous components across hierarchical layers.
- Ensuring **scalability** and **responsiveness** in dynamic environments.

CPS-Controll

CPS Control is the set of methods and principles used to monitor, regulate, and adapt the behavior of Cyber-Physical Systems. It connects real-world processes (e.g., machines, production lines, people flows) with digital decision-making through feedback, models, and intelligent control mechanisms. Rooted in cybernetics and systems theory, CPS Control enables real-time responsiveness and system-level optimization, especially in complex and dynamic environments like manufacturing, logistics, or smart cities.

Core Elements CPS-Controll :

- **Control & Regulation (Cybernetics):** Based on feedback and information flow, enabling systems to react to changes in the environment.
- **Model-based Control:** High-quality models are essential for understanding behavior and enabling effective decision-making in complex systems.
- **Feedback Loops:** Continuous data from physical processes feeds into the cyber layer, allowing dynamic adjustment.
- **Decision Logic / Policies:** CPS apply rules or learned patterns to trigger appropriate responses to physical states.

Foundational Theorys :

- **Cybernetics:** Explores how systems can be steered and regulated through feedback, communication, and modeling.
- **System Theory:** Helps manage system complexity through modular design and hierarchy.
- **Ashby's Law of Requisite Variety:** A controller must match the variety of the system to manage it effectively.

Cyber Layer Functions :

- **Complex Event Processing (CEP):** Detects relevant patterns in data streams for real-time decision-making.
- **Statistical Process Control (SPC):** Identifies process variations to ensure stability and predictability.
- **Data Integration:** Combines sensor data, historical records, and predictions to drive control actions.

Key-Challenges :

- **System Complexity:** Multiple interrelated components require coordinated control strategies.
- **Data Variety & Quality:** CPS rely on accurate, timely, and integrated data to make reliable decisions.
- **Scalability & Hierarchy:** Control must work across multiple system levels—from sensors to enterprise planning.

- **Explainability & Adaptivity:** Control decisions must be interpretable and adaptable to changing environments.

26. Target-Centric Strategy

A **Target-Centric Strategy** refers to the structured alignment of activities, technologies, and data around specific business goals, subjects, or stakeholder needs. Rather than taking a generic or system-centered approach, it emphasizes a clear focus—whether on enterprise objectives, customer value, or strategic priorities—to guide actions, optimize resource use, and drive measurable outcomes. It is essential for ensuring that systems like Business Intelligence, data platforms, or IT architectures serve concrete, goal-oriented purposes.

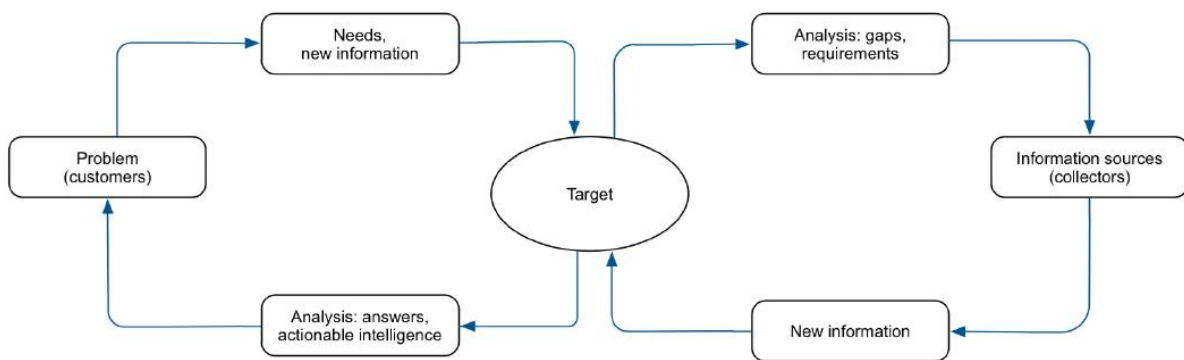


Fig: Intelligence analysis: a target-centric approach by Robert M Clark (Clark, 2013)

Core Characteristics

- **Goal orientation:** Activities are structured around strategic business objectives.
- **Subject orientation:** Systems (e.g., Data Warehouses) are organized around key business subjects like customers or products.
- **Customer focus:** Decision-making and data analysis center on customer needs and profitability.
- **Priority setting:** Resources and actions are allocated based on clear priorities derived from strategy.

Examples from Practice

- **Strategic alignment in IT and BI:**
 - BI systems are built to reflect and support enterprise goals.
 - IT strategies are designed in line with corporate vision and transformation needs.
- **Subject- and theme-based data systems:**

- Data Warehouses and Operational Data Stores are organized by business-relevant themes (e.g., customer, region).
- **Customer-centered strategies:**
 - Design Thinking, segmentation, and campaign management target high-value customers and customize experiences.
 - "Customer 360" views unify data across systems for strategic customer insights.
- **Strategic project prioritization:**
 - Portfolio management balances resources and goals by aligning with measurable KPIs and transformation targets.
- **Data management with business focus:**
 - Data lineage efforts prioritize tracing only business-critical data flows.

Why it Matters

- Promotes **efficient use of data and systems**.
- Enables **informed, goal-driven decision-making**.
- Improves **accountability and performance tracking**.
- Supports **organizational agility** by keeping focus on what matters most.

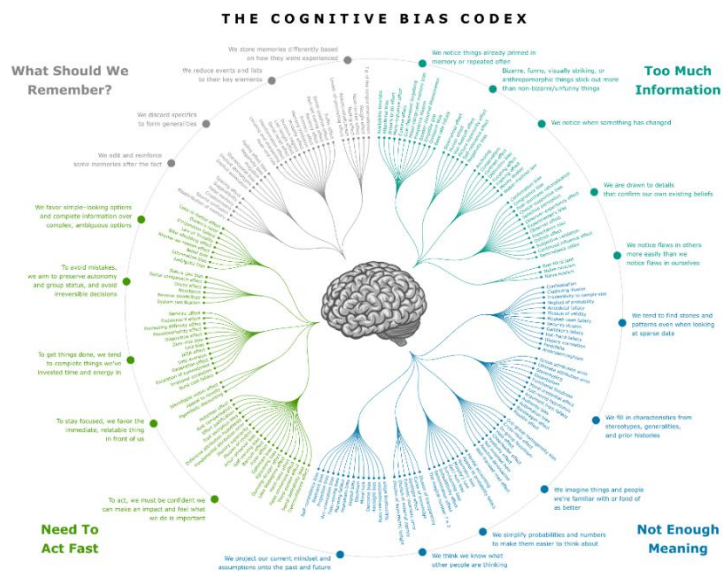
27. Fairness – Risiken

Fairness-Related Risks :

1. Definition & Evaluation
 - Fairness is treated as a standalone quality goal alongside accuracy and transparency.
 - Assessments often rely on the FAT framework (Fairness, Accountability, Transparency).
 - How the data are partitioned—by latent features, demographic attributes, or classes—strongly influences fairness metrics.
 - Empirical studies indicate that fairness results remain relatively stable across different data splits.
2. Stakeholder Perspectives
 - Model developers frequently emphasize predictive accuracy.
 - Business stakeholders also demand fair decision-making.
 - Model consumers must dig into model internals to diagnose robustness and fairness issues.
3. Ethical Considerations
 - Fairness is one of the core ethical principles in data science.

- In learning-analytics settings, data privacy, ownership, transparency, and informed consent are crucial.
- Improper sampling or bias in data collection can lead to systematically unfair treatment of certain groups.

Take-away: Achieving fair models requires clear fairness metrics, alignment of stakeholder priorities, and explicit attention to ethics throughout the entire data-to-model lifecycle.



28. Ensemble Model Benefits

- **Predictive Power:** Combining multiple models typically raises R^2 , meaning ensembles outperform any single model in accuracy and robustness.
- **Mitigation via Aggregation:** By averaging (or otherwise aggregating) the individual predictions, ensembles exploit each model's strengths while canceling out their weaknesses, leading to more stable results.

Goal: Higher Accuracy, better Performance

Core Idea:

Ensemble learning involves assembling multiple models, such as:

- regression,
- KNN,
- decision trees, and
- clustering

into a cohesive unit. The core idea is to leverage diverse algorithms or distinct datasets with the same algorithm to achieve a more reliable and accurate prediction. The collective intelligence of multiple models often outshines individual predictions (Gandhi, 2024).

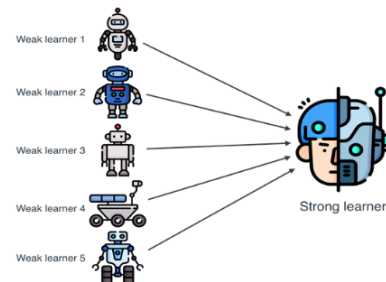


Fig: Ensemble Techniques in Machine Learning (Gandhi, 2024)

29. Evaluation Metrics

Evaluation metrics translate business or research goals into concrete, measurable numbers that tell us whether a data-mining initiative—or any model-driven workflow—delivers value. A robust evaluation plan therefore spans several layers, from goal formulation through model comparison to deployment monitoring.

Data Quality Metrics:

- Quality indicators for learning analytics metrics encompass standards (comparability, consistency), completeness, accuracy (effectiveness, efficiency), validity, timeliness, and uniqueness
- Examples of Data Quality (DQ) metrics include governance and compliance metrics, data quality measurement results, improvement trends, and problem management metrics
- It is crucial to establish methods for measuring adherence to data quality expectations and to define acceptance thresholds for each measurement

Metrics for Data Governance and Stewardship:

- These include compliance with regulatory and internal data policies, value, effectiveness, and sustainability. The Data Mesh framework treats data as a product and emphasizes adherence to organization-wide data governance standards

Performance Evaluation:

- Performance evaluation can be achieved by assigning a measurement scheme using a linking or mapping table that defines what performance should be measured between tracking points and how much time processing is allowed to take
- In the context of Failure Mode and Effects Analysis (FMEA), a Risk Priority Number (RPN) is traditionally calculated by multiplying the significance, probability of occurrence, and detectability of the error [Conversation History]. However, modern approaches sometimes forgo the RPN in favor of risk matrices or tables to define task priorities

30. Data Science History

Data Science — Historical Arc & Key Milestones

1. Conceptual Bedrock

- Data Science builds on the DIKIW hierarchy: *Data* → *Information* → *Knowledge* → *Intelligence* → *Wisdom*.
- Mathematics and statistics provide the language for describing complexity and enable theories to be extended (or reinvented) when traditional tools cannot capture modern data or business problems.

2. Data-Processing Paradigms: ETL vs. ELT

- **ETL (Extract-Transform-Load)**: Classic data-warehouse workflow; data are cleaned and modeled *before* storage, giving highly structured, reporting-ready schemas.
- **ELT (Extract-Load-Transform)**: Favoured in data-lake and data-mesh architectures; raw or semi-structured data are loaded first and shaped *on demand*. This postpones modeling decisions, supports exploratory analytics, and aligns with governance models where each domain team owns its “data product.”

3. Transformation & Quality Pipeline

Filtering, harmonisation (e.g., merging multi-source customer records), aggregation (e.g., hourly summaries of 10-second IoT readings), and enrichment (deriving KPIs) bring disparate data to a common granularity. Each step is logged in metadata to preserve lineage and compliance.

4. Interdisciplinary Fusion

Data Science is a meeting point for statistics, computing, information science, management science, cognitive science, and the social sciences. This blend equips teams to frame problems, build models, and communicate results across technical and executive audiences.

5. From Data Mining to Knowledge Discovery (KDD)

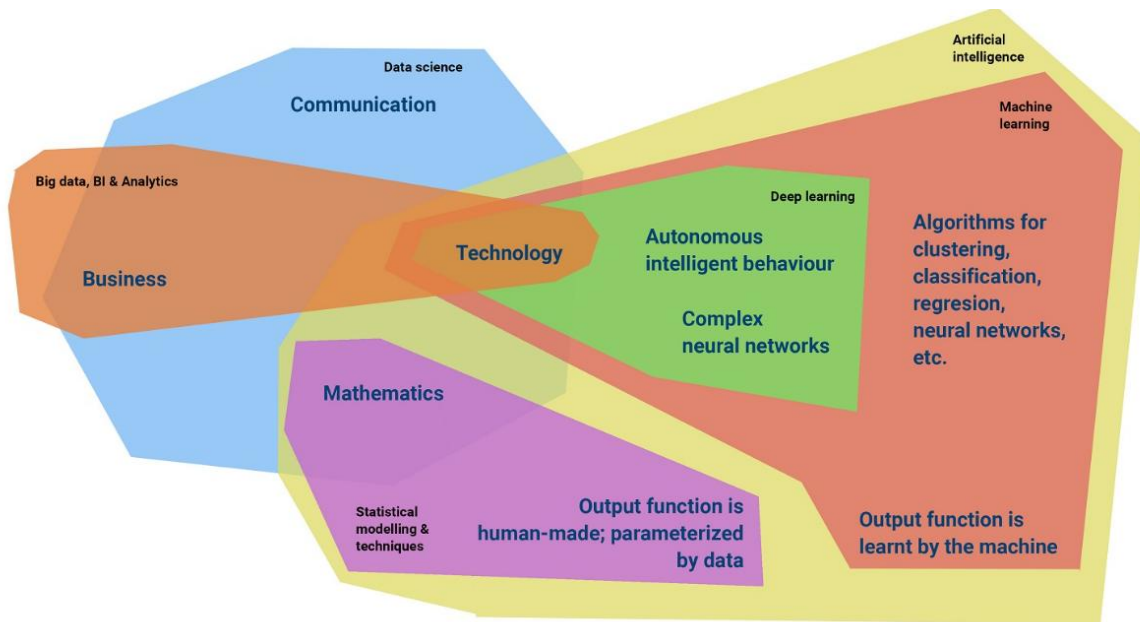
Early “data mining” focused on pattern extraction; KDD broadened the perspective to the entire insight pipeline. Variants such as text, web, and process mining proliferated. Parallel managerial systems evolved:

- **Business Intelligence (BI)**: Enterprise-wide, IT-driven decision support.
- **Management Information Systems (MIS)**: Operational reporting for middle management.
- **Enterprise Information Portals (EIP)**: Aggregated, personalised dashboards that mix internal and external data.

6. Research Frontiers & Modern Themes

Contemporary work spans an A-to-Z of issues—Actionability to Regularisation—highlighting challenges like causality, ethics, heterogeneity, privacy, and quality.

- **Attention Mechanisms:** Transformer models map “query–key” interactions to learn context-dependent meaning.
- **System Dynamics:** Stock-and-flow models (e.g., in STELLA) capture feedback loops in complex systems.
- **Smart Contracts:** Blockchain-based agreements raise questions of legal form, intent, and automated enforcement.



Data Science Roles & How They Interact

